MATH19641: Mathematics 1Q1 Lecture Notes (Completed version) – Weeks 1–6

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Semester 1 – Academic Year 2026/27

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0 Introduction

Welcome to the course notes for Mathematics 1Q1! For a general introduction to the course, including details of assessments, please visit the Course Unit Information module on Canvas.

0.1 Help and support

Course unit lead: Carl Whitfield (you can call me Carl).
E-mail: carl.whitfield@manchester.ac.uk (Start with "Hi Carl").
Office hour: Mondays 10am-11am, Room 1.214 Alan Turing Building

Piazza: Ask questions (anonymously if you wish) and discuss topics with other students on the course in Piazza, accessed via the Canvas page for this course.

Student Support Hub at the Alan Turing Building:
Drop in for Advice, Guidance and Wellbeing Support, Monday to Friday 9:00-17:00

School of Natural Sciences Wellbeing Team: sons.wellbeing@manchester.ac.uk

University Student Support: Central university support and links to the counselling service.

Disability Advisory and Support Service: Register online or book an appointment to get a University Support Plan.

0.2 About the course

The aim of this course is to provide a comprehensive introduction to concepts in algebra and calculus for year 1 Chemistry students. Below you will find the Intended Learning Outcomes (ILOs) for this course unit.

Intended Learning Outcomes

On completion of this course unit, students will be able to

- **[ILO1]** Use equations in implicit or parametric form including straight lines or conic sections to extract relevant information.
- **[ILO2]** Identify the appropriate partial fractions form and find the values of any coefficients for a rational function of a single variable.
- **[ILO3]** Differentiate functions and hence locate and classify stationary points for a function of a single variable.
- **[ILO4]** Recall and apply appropriate rules to integrate functions and use these to measure areas associated with curves or combinations of curves.
- **[ILO5]** Calculate sums, differences and products involving vectors and use these to solve geometrical problems.
- **[ILO6]** Carry out arithmetic using complex numbers and hence solve polynomial and power equations.
- **[ILO7]** Model chemical processes using differential equation systems and solve these systems using a variety of techniques.

0.3 How to use these notes

These notes come in two forms,

- The "gappy notes", which contain gaps in the exercise solutions for you to fill as we work through the material in lectures
- **[This version].** The complete notes, which contains all the full solutions for examples in the lectures and supplementary videos (available on Canvas once the material has been covered in lectures).

In this PDF version of the notes, some of the HTML functionality is lost (such as interactive visualisations), to use these, please see the HTML version in Canvas.

The example exercises in this document are split into three types

- **[Lecture]** These examples will be covered in the lecture.
- **[DIY]** There will be time during lectures to complete these exercises.

- **[Video]** These examples are covered in videos, which can be accessed via Canvas.

0.4 AI use declaration

Generative AI (Microsoft Copilot) has been used to

- Transcribe the course slide deck into a first draft of these course notes, which were subsequently extensively edited manually.
- Generate the javascript scripts for interactive visualisations.
- Cross-check the final PDF version for errors or inconsistencies.

Otherwise, all course unit materials including the mathematical content, examples, diagrams, slides, tutorial questions, online quizzes, *etc.* were created by humans (*i.e.* the current and previous course unit leads).

1 Geometry and Functions

Intended Learning Outcomes

The relevant ILOs for this section are

- **[ILO1]** Use equations in implicit or parametric form including straight lines or conic sections to extract relevant information.
- **[ILO2]** Identify the appropriate partial fractions form and find the values of any coefficients for a rational function of a single variable.

1.1 Equations of lines and circles

Learning Outcomes

On completion of this subsection, students will be able to

- Define geometric properties of straight lines and circles algebraically.
- Solve geometric problems involving lines and circles.

1.1.1 Equation of a straight line

An equation of the form

$$ax + by = p \tag{1.1.1}$$

for variables x and y, and constants a, b and p, represents a straight line in x–y space. The equation can be rearranged into other forms. The form

$$y = mx + c \tag{1.1.2}$$

is particularly useful. Here, m represents the **gradient** of the line and c represents the y-**intercept**, *i.e.* the point $(0, c)$ where the line meets the y-axis (figure 1.1). The animation in figure 1.2 (online version only) allows you to control the gradient and intercept of a straight line with sliders.

Some useful properties of straight lines, as demonstrated in figure 1.3, are as follows

(a) Two lines are parallel if they have the same gradient. For example, $y = x/2 + 3$ and $y = x/2$ are parallel.

(b) Two lines are perpendicular if the product of their gradients is -1. For example, $y = -3x + 5$ and $y = x/3$ are perpendicular. Note the exception is vertical lines, where the gradient is undefined.

(c) A horizontal line has equation $y = c$.

(d) A vertical line has equation $x = d$.

(e) Two lines cross at a point where they have the same value of y for a given value of x.

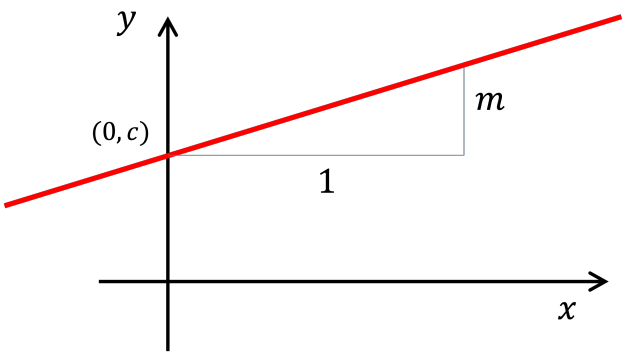


Figure 1.1: Sketch of $y = mx + c$, showing that the y-coordinate of the line increases by m for every distance 1 moved in the x-direction.

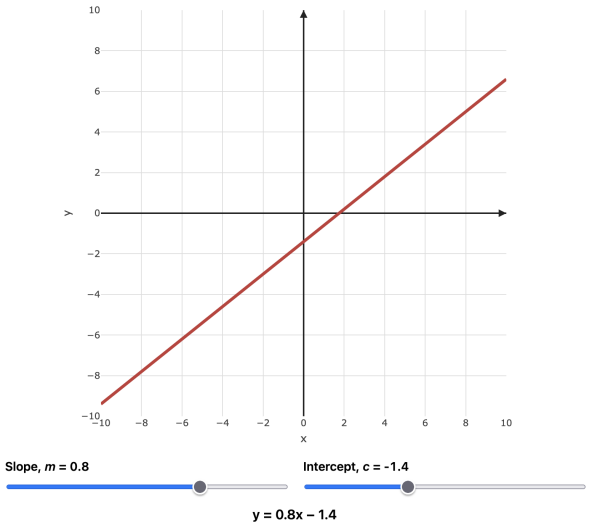


Figure 1.2: Screenshot of the interactive plot of a straight line, available in the HTML version of these notes.

Example 1.1.1 (DIY)**.** Find the crossing point of the lines

$$y = -\frac{1}{2}x + 4, \qquad y = x + 1.$$

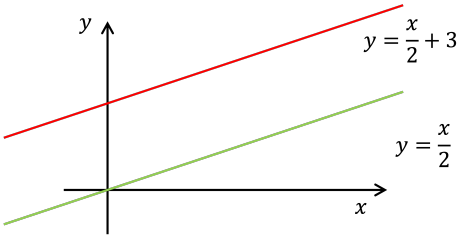
Solution

Equate the two expressions for y:

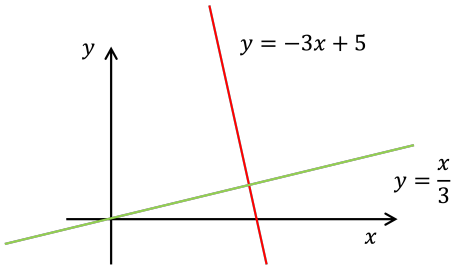
$$-\frac{1}{2}x + 4 = x + 1.$$

Rearranging gives $3 = 3x/2$, and hence $x = 2$. Substitution into the second equation gives $y = 2 + 1 = 3$. Therefore,

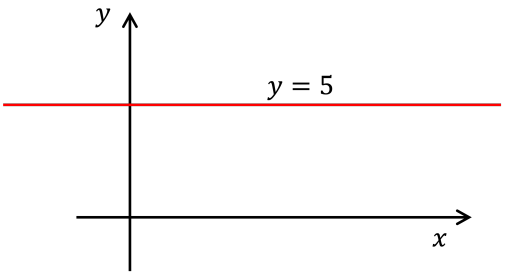
$$(x, y) = (2, 3).$$



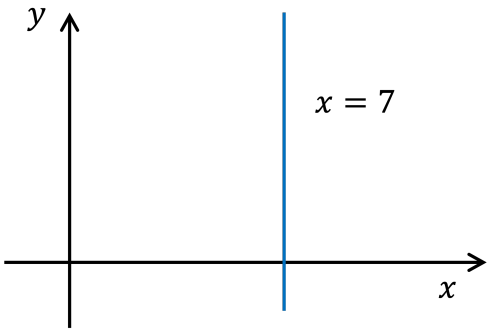
(a) Parallel lines



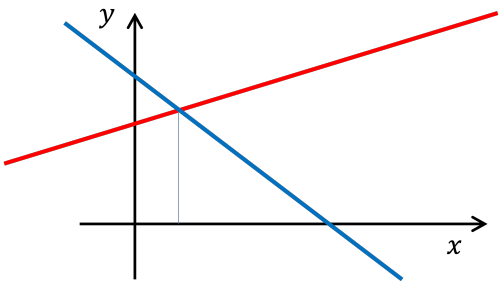
(b) Perpendicular lines



(c) Horizontal line



(d) Vertical line



(e) Two lines that cross

Figure 1.3: Examples of straight line properties and relations.

1.1.2 Equation of a circle

The equation of a circle with centre (a, b) and radius R is

$$(x - a)^2 + (y - b)^2 = R^2 \,. \tag{1.1.3}$$

The straight-line distance between two points (x_1, y_1) and (x_2, y_2) is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \,. \tag{1.1.4}$$

Thus, the radius $R = \sqrt{(x - a)^2 + (y - b)^2}$ is the distance from any point (x, y) on the circle to its centre, which makes sense! (See figure 1.4).

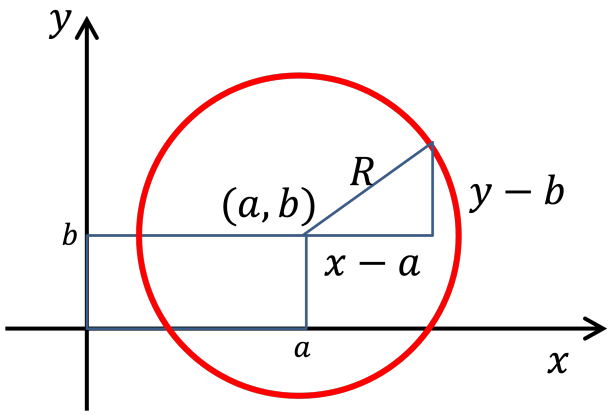


Figure 1.4: Sketch of a circle, with centre (a,b) and radius R, which is also the hypotenuse of the right-angled triangle with side lengths $x-a$ and $y-b$.

Example 1.1.2 (Video)**.** Find the centre and radius of the circle with equation

$$x^2 + 4x + y^2 - 6y = 3.$$

Solution

Complete the square for the quadratics in x and y:

$$x^2 + 4x = (x+2)^2 - 4\,, \qquad y^2 - 6y = (y-3)^2 - 9.$$

Substitute these expressions back into the equation for the circle:

$$(x+2)^2 - 4 + (y-3)^2 - 9 = 3\,,$$

so

$$(x+2)^2 + (y-3)^2 = 16\,.$$

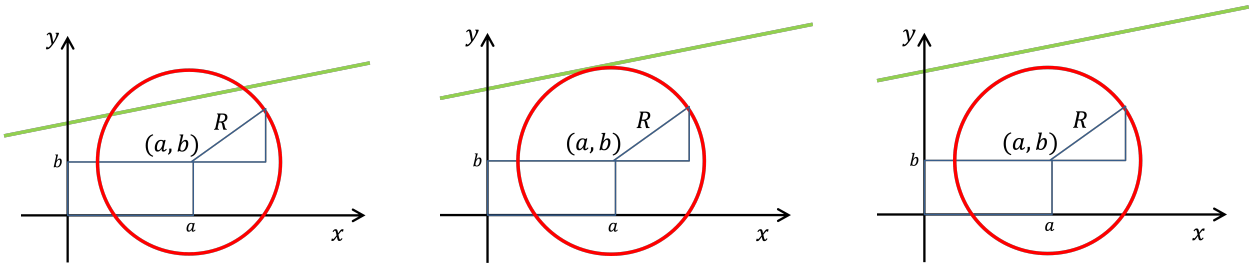
Recall that the equation of a circle with centre (a,b) and radius R is $(x-a)^2 + (y-b)^2 = R^2$. Therefore,

$$a = -2\,, \qquad b = 3, \qquad R = 4.$$

To find where a line and a circle cross, substitute the equation of the line, $y = mx+c$, into the equation of the circle, $(x-a)^2 + (y-b)^2 = R^2$. This gives a quadratic equation for x, which may yield

- two solutions: the line and circle cross twice,
- one repeated solution: the line is tangent to the circle,
- no solutions: line and circle do not cross.

These possibilities are shown in figure 1.5.



(a) Two crossing points (b) One crossing point (c) No crossing points

Figure 1.5: Three possibilities for the crossing points of a line and a circle.

Example 1.1.3 (Lecture). Do the following line and circle cross? If so, at what point or points?

$$y = x + 1, \qquad (x-1)^2 + (y-2)^2 = 8.$$

Solution

Substitute the line equation into the circle and rearrange,

$$(x-1)^2 + (x+1-2)^2 = 8$$
$$\therefore 2(x-1)^2 = 8\,.$$

Solve the resulting quadratic

$$x - 1 = \pm 2$$
$$\therefore x = \{-1,\ 3\}\,.$$

Next, substitute the solutions into the line equation to get

$$y = \{0, 4\}\,,$$

respectively.

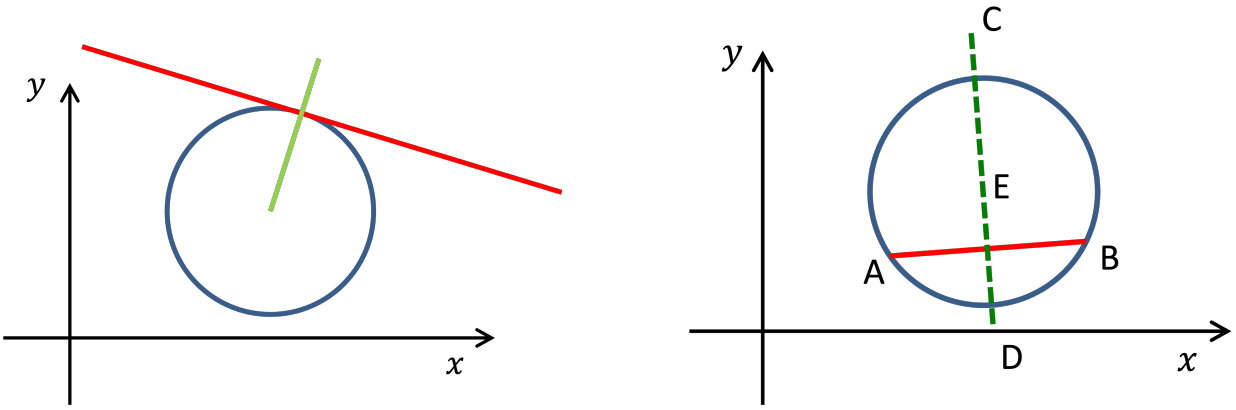
Therefore, the line and circle cross at $(-1, 0)$ and $(3, 4)$.

At any point on a circle, the radius and tangent are perpendicular, as shown in figure 1.6(a).

Useful definition: The **perpendicular bisector** of a line segment AB is a straight line that passes through the midpoint of AB and is perpendicular to it. A straight line segment connecting two points on a circle is a **chord**.

Perpendicular bisector theorem: Given two points A and B, all points on the perpendicular bisector of the line segment AB will be the same distance from A and B. Therefore, if A and B lie on a circle,

the perpendicular bisector of the chord AB will pass through the centre, as shown in figure 1.6(b).



(a) Tangent line

(b) Perpendicular bisector

Figure 1.6: Sketches of (a) the tangent to a circle, which is perpendicular to its radius and (b) the perpendicular bisector (CD) of two points on the circle (A and B), which passes through the circle's centre E.

Example 1.1.4 (Lecture)**.** Find the equation of a circle with centre along the line $y = 2x - 3$ and which also passes through $(1, 6)$ and $(5, 2)$.

Solution

Use the perpendicular bisector theorem. The gradient between the points is

$$\frac{\Delta y}{\Delta x} = -1\,.$$

The midpoint is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = (3, 4)\,.$$

Therefore, the equation of the perpendicular bisector is

$$y = x + 1\,.$$

The centre is the crossing point of the perpendicular bisector with $y = 2x - 3$:

$$(a, b) = (4, 5)\,.$$

To find R, substitute any point on the circle:

$$R^2 = (1 - 4)^2 + (6 - 5)^2 = 10\,.$$

Solution (cont)

Putting everything together gives

$$(x-4)^2+(y-5)^2=10\,.$$

⋆ Exam mistake of the week ⋆

Trying to solve for three variables using only two equations.

> *25/26 Exam Q1(b):* The (x,y) points $A=(2,9)$ and $B=(-5,2)$ lie on a circle with radius R and centre (a,b). Given that the circle also touches, but does not cross, the x-axis find the two possible values of R and the corresponding values of (a,b).

Many students failed to use the information that the circle is tangent to the x-axis and attempted to solve by substituting the two given points into the circle equation, which is not sufficient to solve three unknowns (a, b, and R). Students attempted multiple ingenious approaches but this was ultimately time wasted. Note, this problem is Question 4 of Tutorial Sheet 1.

Example 1.1.5 (Video)**.** There are exactly two circles which touch, but do not cross, the x-axis and pass through $A=(0,1)$ and $B=(-2,5)$. Find the centres, radii and hence the equations of these two circles.

Solution

The equation of a circle with centre (a,b) and radius R is given by equation (1.1.3). If the circle touches the x-axis, then $R^2=b^2$ and so

$$(x-a)^2+(y-b)^2=b^2\,.$$

or, expanding out,

$$x^2-2ax+a^2+y^2-2by=0\,.$$

Define $y=mx+c$ as the equation of the perpendicular bisector of AB. The gradient of the line AB is $m_{AB}=(5-1)/(-2-0)=-2$ and so $m=1/2$.

The midpoint of AB is $(X,Y)=((0-2)/2,(1+5)/2)=(-1,3)$, which lies on the bisector, therefore $Y=mX+C$ and so

$$c=Y-mX=3+1/2=7/2$$

Thus, the perpendicular bisector of AB is

$$y=\frac{1}{2}x+\frac{7}{2}\,,$$

Solution (cont)

and so $b = a/2 + 7/2$. Finally, any point on the circle, say $A = (0, 1)$, must satisfy the equation $x^2 - 2ax + a^2 + y^2 - 2by = 0$. Substituting in $x = 0$, $y = 1$ gives

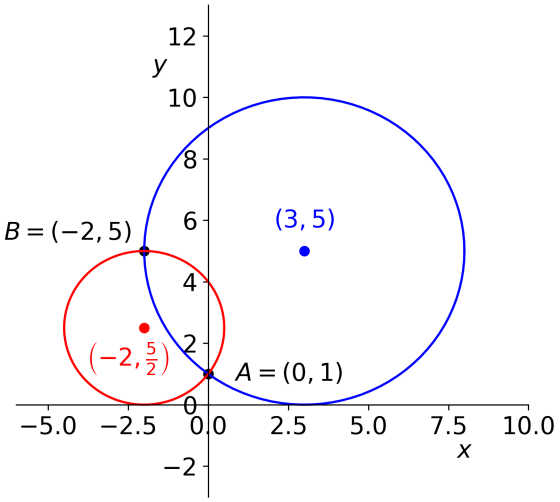
$$a^2 - a - 6 = 0\,.$$

Thus, $a = 3$ or $a = -2$, with corresponding values $b = 5$ and $b = 5/2$ respectively. Therefore, the two circles are given by the equations

$$(x - 3)^2 + (y - 5)^2 = 25\,,$$

and

$$(x + 2)^2 + \left(y - \frac{5}{2}\right)^2 = \frac{25}{4}.$$



Recap

Key points from section 1.1

- **Straight line equation:** $ax + by = p$ or $y = mx + c$.
- Equation of a **circle**: $(x - a)^2 + (y - b)^2 = R^2$.
- **Crossing point** $\rightarrow$ substitute the equation for the line into that of the circle.
- At a point on the circle, the radius and tangent are **perpendicular**.
- Given two points on the circle, an equation for a straight line **passing through the circle's centre** can be found.

1.2 Explicit, implicit and parametric relations

Learning Outcomes

On completion of this subsection, students will be able to

- Recognise when relations between variables are in explicit, implicit or parametric form
- Reformulate explicit equations into parametric and implicit forms.

1.2.1 Implicit relations

An explicit relation expresses the **dependent variable** (*e.g.* y) as a function of the **independent variable** (*e.g.* x), and so has the form $y = f(x)$. An example we have already met is the straight-line equation $y = mx + c$.

An**implicit** relation has the form

$$f(x, y) = d\,, \tag{1.2.1}$$

for some constant d. For a given value of x, there may be multiple y values that satisfy this (and vice versa). An example we have already met is the equation of a circle $(x - a)^2 + (y - b)^2 = R^2$.

In some cases, it is not possible to write a relation explicitly as $y = f(x)$. For example, $4x + 8y + \sin(\pi x) + \cos(\pi y) = 1$, which is plotted in figure 1.7, behaves like an explicit function, with one value of y for each x, however, we cannot reformulate it as an explicit function.

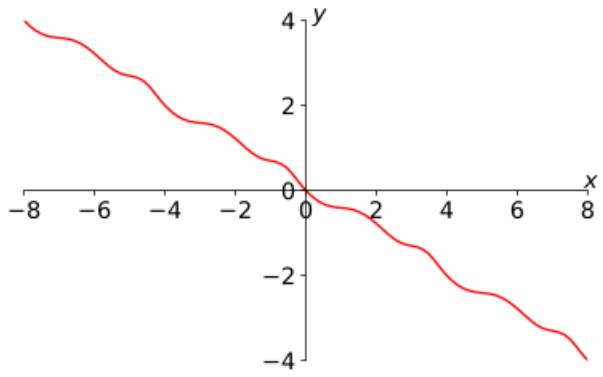


Figure 1.7: A plot of the implicit function $4x + 8y + \sin(\pi x) + \cos(\pi y) = 1$.

Example 1.2.1 (Video)**.** An ellipse can be written in implicit form as

$$\frac{x^2}{100} + \frac{y^2}{25} = 1.$$

For $x = -10, -8, -6, 0, 6, 8, 10$, find the possible values of y and hence draw the ellipse.

Solution

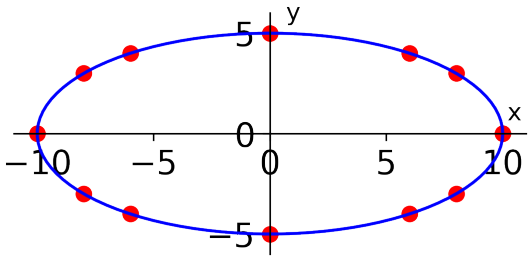
Rearrange to get y in terms of x:

$$\frac{y^2}{25} = 1 - \frac{x^2}{100}$$

$$y = \pm 5\sqrt{1 - \frac{x^2}{100}} = \pm\frac{1}{2}\sqrt{100 - x^2}$$

Substitution gives

x	y
-10	0
-8	± 3
-6	± 4
0	± 5
6	± 4
8	± 3
10	0



1.2.2 Parametric relations

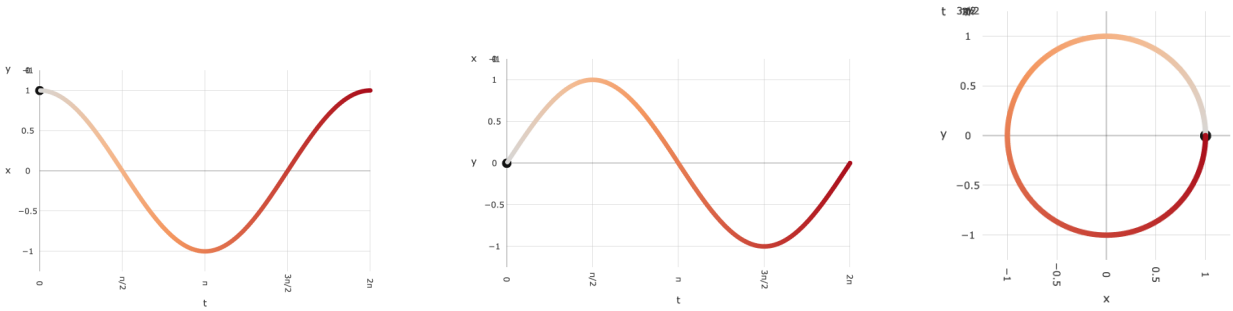
A **parametric relation** between x and y has the form

$$x = f(t), \qquad y = g(t). \tag{1.2.2}$$

So x and y are both functions of the parameter t. For example, the equation of a circle in parametric form is given by

$$x = a + R\cos t, \qquad y = b + R\sin t,$$

for $0 \leq t < 2\pi$. The interactive animation in figure 1.8 demonstrates how parametric relations can be visualised by making the parameter t the coordinate of the third dimension.



(a) $x - t$ plane (b) $y - t$ plane (c) $x - y$ plane

Figure 1.8: Screenshots of the interactive plot of the parametric curve $x = \cos t$, $y = \sin t$, available in the html version of these notes. The colour goes from white to red as t increases.

Example 1.2.2 (DIY)**.** Express the circle with centre at $(2, 3)$ and radius 4 in parametric form.

Solution

In parametric form

$$x = a + R\cos t, \qquad y = b + R\sin t.$$

For $a = 2$, $b = 3$, and $R = 4$, this gives

$$x = 2 + 4\cos t, \qquad y = 3 + 4\sin t,$$

for $0 \leq t < 2\pi$.

Given $x = f(t)$ and $y = g(t)$, it may or may not be possible to express y directly as a function of x. For example, consider the parametric relation $x = t^2$, $y = t^4$. It can be written as $y = x^2$. However, for all values of t, $x \geq 0$ (see figure 1.9).

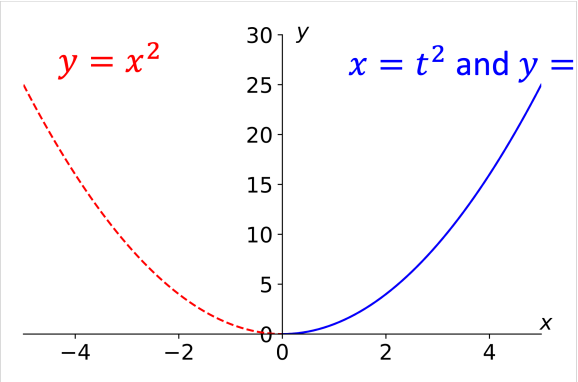


Figure 1.9: Comparison of the function $y = x^2$ and the parametric relation $x = t^2$, $y = t^4$.

A general method to write parametric equations in explicit form is:

- rearrange the equation for the **independent variable** in terms of the parameter, i.e. rewrite $x = f(t)$ into $t = f^{-1}(x)$;
- substitute this into the equation for the **dependent variable**, i.e. replace t with $f^{-1}(x)$ in $y = g(t)$.

Note: $t = f^{-1}(x)$ might not be unique, so be careful not to lose valid solutions! (*E.g.* $x = t^2$ means $t = \pm\sqrt{x}$).

Example 1.2.3 (Lecture)**.** Sodium oxide ($\mathrm{Na_2O}$) is placed in water producing sodium hydroxide:

$$\mathrm{Na_2O} + \mathrm{H_2O} \longrightarrow 2\mathrm{NaOH}.$$

The number of $\mathrm{Na_2O}$ and NaOH molecules in solution at time $t \geq 0$ is given by

$$x = x_0 e^{-\alpha t} \qquad \text{and} \qquad y = 2x_0\left(1 - e^{-\alpha t}\right)$$

respectively, where x_0 is the initial number of molecules of $\mathrm{Na_2O}$. Write y as a function of x, noting any restrictions on their values.

Solution

Find substitution for t in terms of x:

$$e^{-\alpha t} = \frac{x}{x_0}.$$

Substitute into equation for y:

$$y = 2(x_0 - x).$$

x is a decreasing function of t, since $t \geq 0$, the minimum and maximum x values are

$$0 < x \leq x_0.$$

y is an increasing function of t, the minimum and maximum y values are

$$0 \leq y < 2x_0.$$

The straight line and the circle can both be expressed in parametric and implicit form.

Straight Line

- Implicit: $ax + by = p$.
- Parametric: $x = bt$, $y = \frac{p}{b} - at$.
- Explicit: $y = -\frac{a}{b}x + \frac{p}{b} = mx + c$.

Circle

- Implicit: $(x-a)^2+(y-b)^2=R^2$.
- Parametric: $x=a+R\cos t$, $y=b+R\sin t$.
- Explicit: $y=b\pm\sqrt{R^2-(x-a)^2}$.

Recap

Key points from section 1.2

- An **explicit relation** expresses the dependent variable as a function of the independent variable (e.g. $y=f(x)$).
- An **implicit relation** expresses the relationship between the variables via a function of both, (e.g. $f(x,y)=0$).
- A **parametric relation** expresses the variables as functions of a **parameter** (e.g. $x=f(t)$ and $y=g(t)$).

1.3 Partial fractions

Learning Outcomes

On completion of this subsection, students will be able to

- Simplify proper and improper compound fractions into partial fractions form

1.3.1 Partial fractions in general

What do we mean by "partial fractions"? Consider the following sum of two fractions:

$$\frac{1}{x+1}+\frac{3}{x-2}.$$

We can put them over a common denominator to express as a single fraction (see figure 1.10):

$$\frac{4x+1}{(x+1)(x-2)}.$$

Partial fraction decomposition is about **reversing** this operation. This is often useful for differentiation and integration.

$$\frac{4x+1}{(x+1)(x-2)} = \frac{1}{x+1} + \frac{3}{x-2}$$

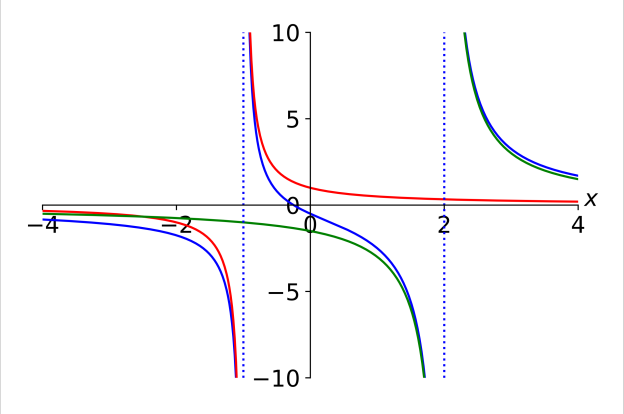


Figure 1.10: Plots of the irreducible fractions $\frac{1}{x+1}$ + and $\frac{3}{x-2}$ and their sum, equivalent to $\frac{4x+1}{(x+1)(x-2)}$.

Useful definitions:

- A **rational fraction** has a numerator and denominator that are both polynomials.
- The **degree** of a polynomial is the largest power in the polynomial, *e.g.* $x^3 + x^2 - x + 1 = 0$ is degree 3.
- Defining d as the degree of the denominator and n as the degree of the numerator, then a **proper fraction** is a rational fraction where $d > n$. An **improper fraction** is a rational fraction where $d \leq n$.

1.3.2 Partial fractions for proper fractions

Case 1: Linear function divided by factorising quadratic. A linear function divided by a factorising quadratic can be written in the form

$$\frac{\text{linear}}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}\,.$$

To find A and B, first multiply both sides by the denominator $(x-a)(x-b)$ to yield

$$\text{linear} = A(x-b) + B(x-a).$$

This equation must hold **for all** x. So, set $x = a$ to find A (in terms of a, b, and the linear function) and set $x = b$ to find B (in terms of a, b, and the linear function).

Example 1.3.1 (Lecture)**.** Express in partial fractions form

$$\frac{x-4}{2x^2+5x-3}.$$

Solution

Factorise the denominator

$$2x^2 + 5x - 3 = (x + 3)(2x - 1).$$

Write the form of the solution

$$\frac{x-4}{(x+3)(2x-1)} = \frac{A}{x+3} + \frac{B}{2x-1}.$$

Multiply through by the denominator

$$x - 4 = A(2x - 1) + B(x + 3).$$

Solve for A and B

Substituting $x = -3$ gives $-7 = -7A$, therefore $A = 1$

Substituting $x = 1/2$ gives $-7/2 = -7B/2$, therefore $B = -1$

and so

$$\frac{x-4}{2x^2+5x-3} = \frac{1}{x+3} - \frac{1}{2x-1}.$$

Case 2: Quadratic function divided by a factorising cubic. A quadratic function divided by a factorising cubic can be written in the form

$$\frac{\text{quadratic}}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}.$$

To find A, B and C, multiply through by the denominator:

$$\text{quadratic} = A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b).$$

Then set $x = a$ to find A, $x = b$ to find B, and $x = c$ to find C.

Case 1 and 2 generalise to any proper fraction with linear factors on the denominator:

$$\frac{\text{degree } n-1 \text{ polynomial}}{(x-x_1)(x-x_2)\cdots(x-x_n)} = \frac{A_1}{x-x_1} + \frac{A_2}{x-x_2} + \cdots + \frac{A_n}{x-x_n}. \tag{1.3.1}$$

Example 1.3.2 (Video)**.** Express in partial fractions form

$$\frac{2x^2+8x+18}{x(x+2)(x-3)}.$$

Solution

The denominator contains 3 linear factors, so the correct form is

$$\frac{2x^2+8x+18}{x(x+2)(x-3)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-3}.$$

Multiplying both sides by the denominator:

$$2x^2+8x+18 = A(x+2)(x-3) + Bx(x-3) + Cx(x+2).$$

This must hold for all x, so substitute $x=0$ to cancel B and C terms:

$$18 = A(2)(-3) = -6A \qquad \therefore\ A = -3.$$

Substitute $x=-2$ to cancel A and C terms:

$$8-16+18 = B(-2)(-5) \qquad \therefore\ 10B = 10 \qquad \therefore\ B = 1.$$

Substitute $x=3$ to cancel A and B terms:

$$18+24+18 = C(3)(5) \qquad \therefore\ 15C = 60 \qquad \therefore\ C = 4.$$

And so

$$\frac{2x^2+8x+18}{x(x+2)(x-3)} = -\frac{3}{x} + \frac{1}{x+2} + \frac{4}{x-3}.$$

Case 3: Repeated factor in the denominator. A repeated factor in the denominator results in two fractions, one with a square on the denominator, e.g.

$$\frac{\text{quadratic}}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}\,.$$

Multiplying both sides by the denominator:

$$\text{quadratic} = A(x-a)(x-b) + B(x-b) + C(x-a)^2.$$

Setting $x=a$ and $x=b$ only gives two of A, B, C. It is necessary to find the third by comparing coefficients of a power of x (e.g. comparing constants by setting $x=0$).

This case also generalises to factors repeated n times, e.g.

$$\frac{\text{degree } n \text{ polynomial}}{(x-a)(x-b)^n} = \frac{A}{x-a} + \frac{B_1}{x-b} + \cdots + \frac{B_n}{(x-b)^n}. \tag{1.3.2}$$

Example 1.3.3 (DIY)**.** Write down the partial fractions form for

$$\frac{x+9}{x(x^2-6x+9)},$$

but **do not** find the values of the coefficients.

Solution

Factorise the denominator:

$$x^2-6x+9=(x-3)^2.$$

Three terms required for x, $x-3$, and $(x-3)^2$. Therefore, the partial fraction form is

$$\frac{x+9}{x(x^2-6x+9)}=\frac{A}{x}+\frac{B}{x-3}+\frac{C}{(x-3)^2}.$$

Case 4: Irreducible quadratic in the denominator. An irreducible quadratic factor in the denominator results in a term of the form $\frac{Ax+B}{\text{quadratic}}$, *e.g.*

$$\frac{\text{quadratic}}{(x^2+2x+9)(x-1)}=\frac{Ax+B}{x^2+2x+9}+\frac{C}{x-1}\,.$$

Once again, it is necessary to use comparison of coefficients in order to find some of the values A, B, C etc.

This rule generalises to irreducible polynomials of degree n, e.g.

$$\frac{\text{degree } n \text{ polynomial}}{(x-a)(\text{irreducible degree } n \text{ polynomial})}=\frac{A}{x-a}+\frac{B_{n-1}x^{n-1}+\cdots+B_1x+B_0}{\text{irreducible degree } n \text{ polynomial}}. \tag{1.3.3}$$

Note: Here, we will take "irreducible" to mean that it cannot be factorised into integer roots. But, in general, the method works for all roots. *E.g.* In the following example, the quadratic has irrational solutions, and could be factorised using these, but you will not be asked to do this in this course.

Example 1.3.4 (Video)**.** Express in partial fractions form

$$\frac{x^2}{(x-1)(x^2+3x-3)}.$$

Solution

The denominator cannot be factorised further, so the correct form is

$$\frac{x^2}{(x-1)(x^2+3x-3)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+3x-3}.$$

Multiplying both sides by the denominator and collecting powers of x:

$$x^2 = A(x^2+3x-3) + (Bx+C)(x-1)$$
$$x^2 = (A+B)x^2 + (3A-B+C)x - 3A - C\,.$$

Compare coefficients of powers of x:

$$x^0: \quad 0 = -3A - C \qquad \therefore C = -3A,$$

$$x: \quad 0 = 3A - B + C \qquad \therefore B = 0,$$

$$x^2: \quad 1 = A + B \qquad A = 1 \qquad \therefore C = -3.$$

And so

$$\frac{x^2}{(x-1)(x^2+3x-3)} = \frac{1}{x-1} - \frac{3}{x^2+3x-3}.$$

1.3.3 Partial fractions for improper fractions

If a rational function has a denominator of degree n and a numerator of degree $n+m$, where $m \geq 0$ then the resultant partial fraction will include a polynomial of degree m, for example

$$\frac{3x^3+x^2-x+6}{(x-2)(x+1)} = Ax + B + \frac{C}{x-2} + \frac{D}{x+1}$$

To find the coefficients, multiply both sides by the denominator and either substitute meaningful values of x or compare coefficients of powers of x.

Example 1.3.5 (Lecture)**.** Find the partial fractions form of

$$\frac{3(x+1)^2}{x^2+x-2}.$$

Solution

Factorise the denominator:

$$x^2 + x - 2 = (x-1)(x+2).$$

Solution (cont)

Identify correct form:

$$\frac{3(x+1)^2}{(x-1)(x+2)} = A + \frac{B}{x-1} + \frac{C}{x+2}.$$

Multiply through and solve for coefficients:

$$3(x+1)^2 = A(x+2)(x-1) + B(x+2) + C(x-1)$$
$$3x^2 + 6x + 3 = Ax^2 + (A+B+C)x - 2A + 2B - C\,.$$

Compare coefficients of powers of x:

$$x^2: \quad A = 3\,,$$

$$x: \quad A + B + C = 6 \qquad \therefore B + C = 3,$$

$$x^0: \quad -2A + 2B - C = 3 \qquad \therefore 2B - C = 9 \quad \therefore B = 4\,,\, C = -1.$$

Putting it all together

$$\frac{3(x+1)^2}{x^2+x-2} = 3 + \frac{4}{x-1} - \frac{1}{x+2}.$$

Example 1.3.6 (Video)**.** Find the partial fractions form for

$$\frac{5x^6 - 50x^5 + 386x^4 - 1074x^3 + 644x^2 + 1174x - 1354}{(x-2)^2(x^2-6x+50)},$$

but **do not** find the values of the coefficients.

Solution

Check the degree of the numerator (n) and denominator (d):

$$n = 6 \qquad \text{and} \qquad d = 4 \qquad \text{and so} \qquad n - d = 2.$$

Therefore the partial fractions form contains a quadratic.

Check if the denominator factorises further

It does not.

Write the terms for the repeated root and the quadratic:

$$Ax^2 + Bx + C + \frac{D}{x-2} + \frac{E}{(x-2)^2} + \frac{Fx+G}{x^2-6x+50}.$$

Recap

Key points from section 1.3

- Proper fractions ($d > n$):
  - Linear factors in denominator $\rightarrow \frac{A}{\text{linear}}$
  - Twice repeated linear factors $\rightarrow \frac{A}{\text{linear}} + \frac{B}{\text{linear}^2}$
  - Quadratic factors in denominator $\rightarrow \frac{Ax+B}{\text{quadratic}}$
- Improper fractions ($d \leq n$): Add a polynomial of degree $n - d$
- **General procedure:**
  1. Factorise the denominator
  2. Classify the fraction and select the right form
  3. Multiply both sides by the denominator
  4. Compare coefficients (or choose values of x) to solve for the constants

2 Differentiation

Intended Learning Outcomes

The relevant ILOs for this section are

- **[ILO3]** Differentiate functions and hence locate and classify stationary points for a function of a single variable.

2.1 Maxima and minima of functions

Learning Outcomes

On completion of this subsection, students will be able to

- Calculate the derivative of basic mathematical functions
- Identify local and global maxima and minima of a function

2.1.1 Reminder of basic differentiation

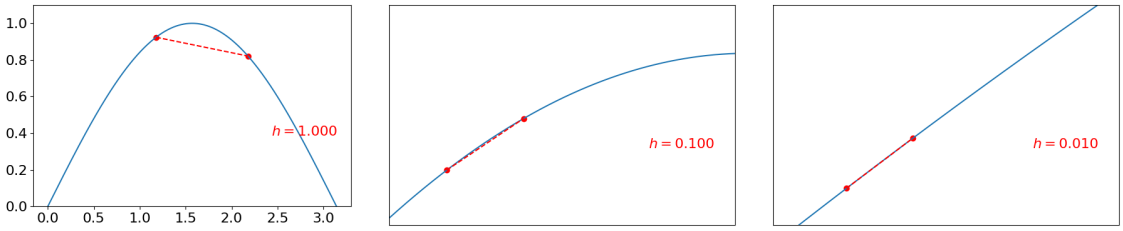
What is the derivative? We have calculated the gradient of a straight line as

$$m = \frac{\Delta y}{\Delta x}.$$

The derivative of a function $f(x)$ is the gradient at a point (i.e. as the change in x gets infinitesimally small)

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}. \tag{2.1.1}$$

Another way to imagine this is demonstrated in figure 2.1. As we "zoom in" on any smooth function, it looks more like a straight line with gradient equivalent to the tangent line at that point, *i.e.* the derivative. Some common derivatives of functions are given in table 1.



(a) Zoomed out, $h = 1$. (b) Zooming in, $h = 0.1$. (c) Zoomed in, $h = 0.01$.

Figure 2.1: The straight line between two points on the curve $\sin x$ gets closer to a tangent to the curve as we zoom in and reduce h, becoming a better approximation of the derivative at the left-hand point, which remains fixed.

Function $y = f(x)$	Derivative $\mathrm{d}y/\mathrm{d}x = f'(x)$
x^n	nx^{n-1}
$\ln x$	$\dfrac{1}{x}$
e^x	e^x
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\sin^{-1} x$	$\dfrac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\dfrac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\dfrac{1}{1+x^2}$

Table 1: Some common functions and their derivatives (see the mathematical formula tables for more standard derivatives).

The second derivative. When a function is differentiated, the resulting derivative is also a function. It can, itself be differentiated to form the second derivative, denoted as

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} \qquad \text{or} \qquad f''(x)$$

2.1.2 Local maxima and minima

Stationary points. A point where $f'(x) = 0$ is called a stationary point. It may represent a place where the function is a maximum or a minimum, but it could represent something else (see figure 2.2). To find stationary points, we need to solve $f'(x) = 0$ for x, there could be

- no stationary points
- one stationary point
- several stationary points.

The first derivative test. To use this test to classify a stationary point at $x = x_0$,

1. evaluate $f'(x)$ at two points either side of x_0, *i.e.* $f'(x_0 - h)$ and $f'(x_0 + h)$ for small h. Make sure h is small enough so that $x_0 \pm h$ does not cross any other stationary points.
2. Drawing the line \for a negative value, / for a positive value and – for the stationary point itself, sketch how the function looks around the stationary point (*e.g.* a minimum will be \– /, and a maximum / –\).

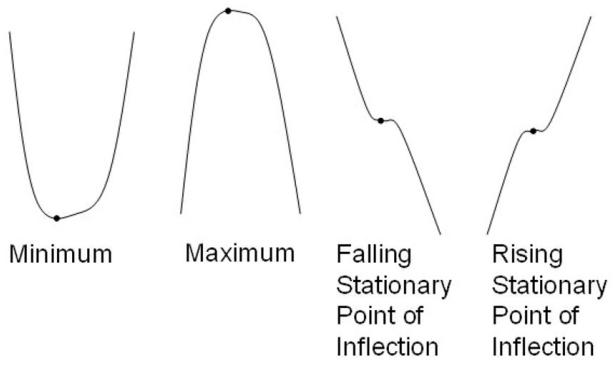


Figure 2.2: Sketch of the four types of stationary point.

The second derivative test. To use this test to classify a stationary point at $x = x_0$, evaluate the second derivative $f''(x)$ at $x = x_0$.

- If $f''(x_0) > 0$, the point is a minimum.
- If $f''(x_0) < 0$, the point is a maximum.
- If $f''(x_0) = 0$, the test is inconclusive. Fall back on the first derivative test.

Example 2.1.1 (Lecture)**.** Find any stationary points for the function

$$f(x) = 2x^3 - 9x^2 + 5\,.$$

Use the first derivative test to classify them as maxima, minima or points of inflection. Check this using the second derivative test.

Solution

Find the first derivative

$$f'(x) = 6x^2 - 18x.$$

Solve for $f'(x) = 0$. Stationary points are at

$$x = 0 \qquad \text{and} \qquad x = 3.$$

First derivative test for $x = 0$:

x	-1	0	1
$f'(x)$	24	0	-12
sketch	/	–	\

Solution (cont)

First derivative test for $x = 3$:

x	2	3	4
$f'(x)$	-12	0	24
sketch	\	–	/

Find the second derivative:

$$f''(x) = 12x - 18.$$

$$f''(0) = -18, \qquad f''(3) = 18.$$

Hence $x = 0$ is a maximum and $x = 3$ is a minimum.

2.1.3 Maximum and minimum values over an interval

Consider the function $f(x)$ in figure 2.3(a); where is the maximum? Where is the minimum? We can see there are stationary points at a (**local** maximum) and b (**local** minimum), but these are not the greatest or smallest values of $f(x)$ over the interval shown. In fact, the minimum over the interval shown is at the left boundary, and the maximum at the boundary.

The greatest/least values of a function within an interval can occur at the following points (labelled on figure 2.3(b))

(A) Stationary points (local maxima/minima),

(B) Ends of the interval for x,

(C) Where $f(x)$ becomes infinite for finite x,

(D) Where $f(x)$ or $f'(x)$ does not exist,

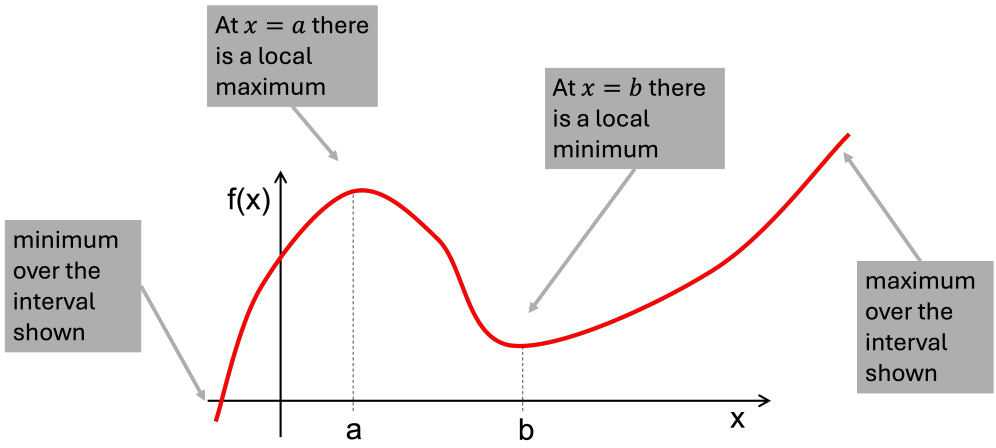
(E) As x becomes infinite (positive or negative).

Such points are often referred to as **critical points**. It is a case of checking all such points in the interval.

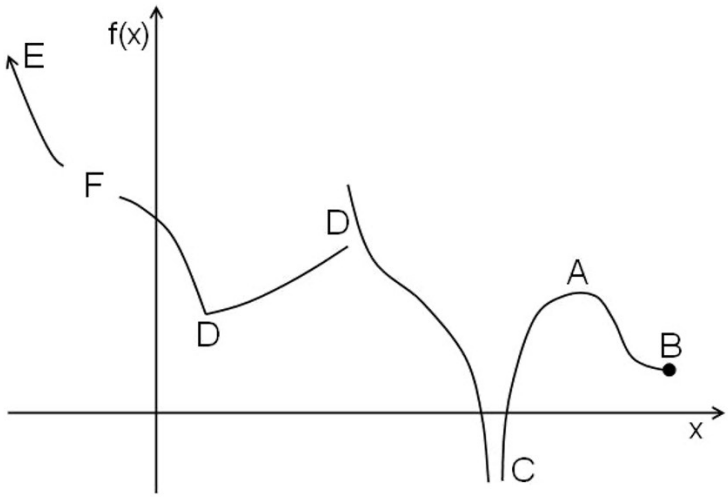
Example 2.1.2 (Lecture)**.** Find the maximum and minimum values of the function

$$p(x) = 2x^3 - 3x^2 - 36x + 13$$

on the interval $-4 \leq x \leq 6$.



(a) A function with local minima/maxima and greatest/least values labelled.



(b) Sketch showing the different candidate points for the greatest/least values of a function in an interval.

Figure 2.3

Solution

Find first derivative

$$p'(x) = 6x^2 - 6x - 36.$$

Solve for $p'(x) = 0$.

$$6(x^2 - x - 6) = 0$$
$$(x - 3)(x + 2) = 0$$

Stationary points are at $x = -2$ and $x = 3$.

Solution (cont)

Values at stationary points:

$$p(-2) = 57, \qquad p(3) = -68.$$

The function $p(x)$ is continuous and finite and the bounds are finite. So just need to check the edges of the domain:

$$p(-4) = -19, \qquad p(6) = 121.$$

Maximum value of 121 at $x = 6$. Minimum value of -68 at $x = 3$.

Example 2.1.3 (Video)**.** A bioreactor produces biomass at a rate of roughly

$$b(\omega) = 3 + \omega \quad [\mathrm{kg/h}],$$

which increases with the stirring rate $\omega \geq 0$ $[\mathrm{rpm}]$. The cost of running the reactor also increases with ω, roughly given by

$$c(\omega) = 72 + \frac{1}{3}\omega^2 \quad [£/\mathrm{h}].$$

Find the stirring rate that optimises production efficiency (i.e. kg of biomass per £ spent), and the associated production efficiency. **Note:** This can be completed once we have covered the quotient rule in section 2.2.3).

Solution

The production efficiency (in kg of biomass per £ spent) is given by

$$f(\omega) = \frac{b(\omega)}{c(\omega)} = \frac{3 + \omega}{72 + \frac{1}{3}\omega^2}.$$

The optimal efficiency will be achieved when $f(\omega)$ is a maximum. Find the stationary points $f'(\omega) = 0$, using the quotient rule:

$$f'(\omega) = \frac{\left(72 + \frac{1}{3}\omega^2\right) - (3 + \omega)\frac{2}{3}\omega}{\left(72 + \frac{1}{3}\omega^2\right)^2} = 0.$$

$$72 + \frac{1}{3}\omega^2 - 2\omega - \frac{2}{3}\omega^2 = 0.$$

Rearranging there is a stationary point when

$$\omega^2 + 6\omega - 216 = 0.$$

Solution (cont)

Factorising (or using the quadratic equation)

$$(\omega + 18)(\omega - 12) = 0.$$

Since $\omega \geq 0$, we have an optimal stirring rate of $\omega = 12$ [rpm]. Substituting this back into the equation for f, we get a production efficiency of

$$f = \frac{3 + 12}{72 + 144/3} = \frac{15}{120} = \frac{1}{8} \quad [\text{kg}/\text{£}].$$

We can check this is indeed the maximum over the domain by checking $f(\omega = 0) = \frac{1}{24}$ and also $\lim_{\omega \to \infty} f(\omega) = 0$ are both less than the value at the stationary point.

Recap

Key points from section 2.1

- The **derivative** of a function is its gradient with respect to its variable.
- **Local stationary points** can be found by solving $f'(x) = 0$.
- A stationary point x_0 is a **maximum** if $f''(x_0) < 0$ and **minimum** if $f''(x_0) > 0$.
- If $f''(x_0) = 0$, use the **first derivative test** instead.
- The max and min over an interval **might not be stationary points**.

2.2 The product, quotient and chain rules

Learning Outcomes

On completion of this subsection, students will be able to

- Identify when to use the chain, product, or quotient rules
- Calculate the derivatives of functions using the chain, product, or quotient rules, where appropriate

2.2.1 The chain rule

Consider the function

$$f(x) = (\sin x)^2.$$

This is a combination of two functions i.e. start with x, take the sine and then square it.

$$x \xrightarrow{\text{Take the sine}} \sin x \xrightarrow{\text{Square it}} (\sin x)^2.$$

The inner function is $g(x) = \sin x$ and the outer function is $h(u) = u^2$.

The chain rule tells us how to differentiate a function of a function. If $f(x) = h(u)$ where $u = g(x)$,

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\mathrm{d}h}{\mathrm{d}u} \cdot \frac{\mathrm{d}g}{\mathrm{d}x}\,, \tag{2.2.1}$$

or in alternative notation

$$f'(x) = h'(u)g'(x)\,,$$

i.e. differentiate the outer function and multiply by the derivative of the inner function.

For the function $f(x) = (\sin x)^2$, the inner function is $u = g(x) = \sin x$. The outer function is $h(u) = u^2$. Using equation (2.2.1),

$$f'(x) = h'(u)g'(x) = 2u\cos x.$$

Expressing in terms of x,

$$f'(x) = 2\sin x \cos x.$$

Example 2.2.1 (DIY). Differentiate

$$p(t) = (t^2 + 1)^3$$

using the chain rule.

Solution

$$p'(t) = 6t(t^2 + 1)^2.$$

Let $f(t) = t^2 + 1$, then $p(t) = f(t)^3$. The derivative of $f(t)$ is $f'(t) = 2t$. Using the chain rule

$$p'(t) = 3f'(t)f(t)^2 = 6t(t^2 + 1)^2.$$

Compare the solution in example 2.2.1 to the result by expanding the brackets

$$\begin{aligned} p(t) &= (t^2 + 1)^3 \\ &= (t^4 + 2t^2 + 1)(t^2 + 1) \\ &= t^6 + 3t^4 + 3t^2 + 1 \end{aligned}$$

then, taking the derivative

$$\begin{aligned} p'(t) &= 6t^5 + 12t^3 + 6t \\ &= 6t(t^4 + 2t^2 + 1) \\ &= 6t(t^2 + 1)^2. \end{aligned}$$

Compare this to your solution to example 2.2.1, did you get the same result? Clearly, the chain rule is more efficient.

Repeated chain rule. Note the chain rule can be applied repeatedly. Consider,

$$q(x) = e^{\sin(x^2+3)}.$$

Write $p(x) = \sin(x^2 + 3)$ so that $q(x) = \exp[p(x)]$. Then, using the chain rule

$$q'(x) = p'(x) \exp[p(x)]$$

. To find $p'(x)$ use the chain rule again. Let $f(x) = x^2 + 3$ so $f'(x) = 2x$. Then

$$p'(x) = f'(x) \cos f(x)\,,$$

and so $p'(x) = 2x \cos(x^2 + 3)$. Using the expression for $q'(x)$ we get

$$q'(x) = 2x \cos(x^2 + 3) e^{\sin(x^2+3)}\,.$$

Example 2.2.2 (Video). (a) Find the derivative of

$$y = e^{3(1-g^2(x))}$$

in terms of the unknown function $g(x)$ and its derivative $g'(x)$.

(b) When $g(x) = \cos x$, what is y'?

Solution

(a) First define

$$h(x) = 3(1 - g^2) = 3 - 3g^2,$$

then

$$h'(x) = -6gg'.$$

And so

$$y' = h'e^h = -6gg'e^{3(1-g^2)}.$$

Solution (cont)

(b) If $g(x) = \cos x$ then $g'(x) = -\sin x$ and therefore

$$\begin{aligned} y' &= -6gg'e^{3(1-g^2)} \\ &= 6\sin x \cos x\, e^{3(1-\cos^2 x)} \\ &= 6\sin x \cos x\, e^{3\sin^2 x}. \end{aligned}$$

The Chain Rule and Linear Functions. Consider the following functions and how to differentiate them.

$$f(x) = \sin\left(4x + \frac{\pi}{2}\right)$$

$$g(x) = 5e^{2x-9}$$

$$h(x) = 7\ln(6-2x)$$

$$p(x) = (7x+9)^{10}.$$

Each of these functions can be differentiated using the chain rule *e.g.*

$$f(x) = \sin\left(4x + \frac{\pi}{2}\right)$$

$$\begin{aligned} f'(x) &= \cos\left(4x + \frac{\pi}{2}\right) \times 4 \\ &= 4\cos\left(4x + \frac{\pi}{2}\right). \end{aligned}$$

Each of these functions is of the form

$$f(ax+b)$$

and has derivative

$$af'(ax+b).$$

Example 2.2.3 (Lecture)**.** Differentiate

$$r(x) = 2\cos(5x - \pi)$$

and

$$p(x) = \ln(2x+1).$$

Solution

$$r'(x) = -10\sin(5x - \pi).$$
$$p'(x) = \frac{2}{2x+1}.$$

2.2.2 The product rule

Consider the function

$$f(x) = x\sin x.$$

This is a product of two functions $u(x) = x$ and $v(x) = \sin x$. The product rule enables the differentiation of a product of two functions.

If

$$f(x) = u(x)v(x),$$

then the **product rule** states

$$\frac{\mathrm{d}}{\mathrm{d}x}(uv) = v\frac{\mathrm{d}u}{\mathrm{d}x} + u\frac{\mathrm{d}v}{\mathrm{d}x}. \tag{2.2.2}$$

or, in alternative notation

$$f'(x) = u'(x)v(x) + u(x)v'(x),$$

i.e. multiply each function by the derivative of the other and add together. Note, it **does not matter** which function is u and which function is v.

In the case $f(x) = x\sin x$, let $u(x) = x$ and $v(x) = \sin x$. Then $u'(x) = 1$ and $v'(x) = \cos x$. Using $f'(x) = uv' + vu'$,

$$f'(x) = x\cos x + \sin x.$$

Example 2.2.4 (Lecture)**.** Differentiate

$$g(x) = 3x^2\ln x$$

using the product rule.

Solution

Let $u(x) = 3x^2$ and $v(x) = \ln x$. Then $u'(x) = 6x$ and $v'(x) = 1/x$. Using the product rule $g'(x) = uv' + u'v$,

$$g'(x) = 3x + 6x\ln x = 3x(1 + 2\ln x).$$

The product rule for products of three or more functions. Consider

$$f(x) = u(x)v(x)w(x)$$

and make the substitution

$$s(x) = v(x)w(x)$$

so that

$$f(x) = u(x)s(x).$$

Thus

$$f'(x) = u's + us'\,.$$

Using the product rule for $s(x) = v(x)w(x)$:

$$s'(x) = v'w + vw'\,.$$

Substituting into $f'(x)$:

$$f'(x) = u'vw + u(v'w + vw') \tag{2.2.3}$$
$$= u'vw + uv'w + uvw' \tag{2.2.4}$$

gives the sum of three terms. Each is the product of two functions multiplied by the derivative of the third function.

Example 2.2.5 (Video)**.** Differentiate

$$g(x) = x^2 e^x \sin(\pi x)$$

using the product rule.

Solution

Define $u(x) = x^2$, $v(x) = e^x$, and $w(x) = \sin(\pi x)$, so that $g = uvw$.

The product rule for three functions states that

$$g' = u'vw + uv'w + uvw'.$$

Find the derivatives of u, v, and w:

$$u' = 2x, \qquad v' = e^x, \qquad w' = \pi\cos(\pi x).$$

Solution (cont)

And substitute them into the expression for g':

$$g' = 2xe^x \sin(\pi x) + x^2 e^x \sin(\pi x) + x^2 e^x \pi \cos(\pi x).$$

Which can be factorised as

$$g' = xe^x \left[2\sin(\pi x) + x\sin(\pi x) + \pi x \cos(\pi x)\right]$$

$$= xe^x \left[(x+2)\sin(\pi x) + \pi x \cos(\pi x)\right].$$

2.2.3 The quotient rule

Consider the function

$$f(x) = \frac{\sin x}{x}.$$

This is a quotient of two functions *i.e.* one function divided by another. The quotient rule enables the differentiation of a quotient of two functions.

If

$$f(x) = \frac{u(x)}{v(x)},$$

then the quotient rule states

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u}{v}\right) = \frac{v\dfrac{\mathrm{d}u}{\mathrm{d}x} - u\dfrac{\mathrm{d}v}{\mathrm{d}x}}{v^2}. \tag{2.2.5}$$

or in alternative notation

$$f'(x) = \frac{u'v - uv'}{v^2}.$$

Unlike the product rule, it is now **critical** which is u and which is v.

Example 2.2.6 (Lecture)**.** Differentiate

$$g(x) = \frac{\ln x}{x^2}$$

by two methods:

(a) Using the quotient rule

(b) Writing as $g(x) = x^{-2}\ln x$ and using the product rule.

Solution

(a) Let $u(x) = \ln x$ and $v(x) = x^2$. Then

$$u'(x) = \frac{1}{x} \qquad \text{and} \qquad v'(x) = 2x.$$

Using the quotient rule

$$g'(x) = \frac{u'v - uv'}{v^2} = \frac{x - 2x\ln x}{x^4}$$
$$= \frac{1 - 2\ln x}{x^3}.$$

(b) Now let $u(x) = \ln x$ and $v(x) = x^{-2}$. Then

$$u'(x) = \frac{1}{x} \qquad \text{and} \qquad v'(x) = -2x^{-3}.$$

Using the product rule

$$g'(x) = u'v + uv' = \frac{1}{x}x^{-2} - 2x^{-3}\ln x$$
$$= \frac{1 - 2\log x}{x^3}.$$

Example 2.2.7 (Video)**.** Differentiate x with respect to t, where

$$x = \frac{e^{2t}\sin(t^3)}{t}.$$

Solution

Define $u(t) = e^{2t}\sin(t^3)$ and $v(t) = t$ and find their derivatives.

For $u'(t)$, we need to use the product rule and chain rule:

$$u'(t) = 2e^{2t}\sin(t^3) + e^{2t}3t^2\cos(t^3)$$
$$= e^{2t}\left[2\sin(t^3) + 3t^2\cos(t^3)\right].$$

and

$$v'(t) = 1\,.$$

The quotient rule states

$$x' = \frac{u'v - v'u}{v^2}.$$

Solution (cont)

Substituting in the expressions for u, v, u', and v' gives

$$\begin{aligned} x' &= \frac{e^{2t}\left[2\sin(t^3) + 3t^2\cos(t^3)\right]t - e^{2t}\sin(t^3)}{t^2} \\ &= e^{2t}\left[\frac{2t-1}{t^2}\sin(t^3) + 3t\cos(t^3)\right]. \end{aligned}$$

Recap

Key points from section 2.2

- The **chain rule**: if $f(x) = h(u)$ where $u = g(x)$ then

$$f'(x) = h'(u)g'(x).$$

- The **product rule**: if $f(x) = u(x)v(x)$ then

$$f'(x) = u'v + uv'.$$

- The **quotient rule**: If $f(x) = u(x)/v(x)$ then

$$f'(x) = \frac{u'v - uv'}{v^2}.$$

2.3 Implicit, logarithmic, and parametric differentiation

Learning Outcomes

On completion of this subsection, students will be able to

- Identify when to use implicit, logarithmic, and parametric differentiation
- Calculate the derivatives of functions using implicit, logarithmic, and parametric differentiation, where appropriate

2.3.1 Implicit differentiation

Consider the implicit relation

$$x^2 + y^2 = 25.$$

Although written in implicit form, it can be differentiated. It is important to know which variable we are differentiating with respect to. In this case, we want to find $\mathrm{d}y/\mathrm{d}x$.

Differentiate term-by-term with respect to x:

- The derivative of x^2 is $2x$
- Using the chain rule, the derivative of y^2 is $2y\frac{\mathrm{d}y}{\mathrm{d}x}$
- The derivative of 25 is zero

Therefore, we have

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

Which rearranges to

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}.$$

This is an example of **implicit differentiation**.

The second derivative can also be found by repeating the process. Starting with $x^2 + y^2 = 25$ and differentiating we get

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

- Differentiating $2x$ gives 2
- Using the product rule, differentiating $2y\frac{\mathrm{d}y}{\mathrm{d}x}$ gives $2\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 2y\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$

So, we find

$$2 + 2\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 2y\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$$

or

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}{y}.$$

Substituting in the result from before ($\mathrm{d}y/\mathrm{d}x = -x/y$), this reduces to

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1}{y} - \frac{x^2}{y^3}.$$

Alternatively, we could start with the result for $\mathrm{d}y/\mathrm{d}x$, *i.e.*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}.$$

Use the quotient rule to differentiate, with $u = -x$ and $v = y$, so $u' = -1$ and $v' = y'$:

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{u'v - v'u}{v^2} = \frac{-y + xy'}{y^2}.$$

Substituting in the previous result $y' = -x/y$ gives

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{-y - x^2/y}{y^2} = -\frac{1}{y} - \frac{x^2}{y^3},$$

i.e. the same result as before.

Example 2.3.1 (Lecture)**.** If

$$x^4 + 2xy + y^4 = 1,$$

find the derivatives $\mathrm{d}y/\mathrm{d}x$ and $d^2y/\mathrm{d}x^2$ at the point $x = 1$, $y = 0$.

Solution

Differentiate the equation with respect to x and rearrange for $\mathrm{d}y/\mathrm{d}x$.

$$\begin{aligned} 4x^3 + 2y + 2x\frac{\mathrm{d}y}{\mathrm{d}x} + 4y^3\frac{\mathrm{d}y}{\mathrm{d}x} &= 0 \\ (4y^3 + 2x)\frac{\mathrm{d}y}{\mathrm{d}x} &= -(4x^3 + 2y) \\ \frac{\mathrm{d}y}{\mathrm{d}x} &= -\frac{4x^3 + 2y}{2x + 4y^3} \end{aligned}$$

Differentiate the original expression again and rearrange for $d^2y/\mathrm{d}x^2$.

$$\begin{aligned} 12x^2 + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 2x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 12y^2\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 4y^3\frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= 0 \\ (2x + 4y^3)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= -12y^2\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 - 4\frac{\mathrm{d}y}{\mathrm{d}x} - 12x^2 \\ \therefore\ \frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= -\frac{12x^2 + 4y' + 12y^2(y')^2}{2x + 4y^3}. \end{aligned}$$

At $x = 1$ and $y = 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{4 \times 1^3}{2 \times 1} = -2$$

Using this in the expression for the second derivative, we find that at $x = 1$ and $y = 0$,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{12 \times 1^2 + 4 \times (-2)}{2 \times 1} = -2\,.$$

2.3.2 Logarithmic differentiation

Logarithmic differentiation is a special case of implicit differentiation. It can make derivatives simpler by taking advantage of the properties of logarithms, such as

- Transforming products into sums: $\ln(ab) = \ln a + \ln b$

- Transforming powers into coefficients: $\ln(x^n) = n \ln x$

E.g. consider a function of the form

$$y = f(x)g(x)h(x).$$

On taking the logarithm of the absolute value of both sides,

$$\ln|y| = \ln|fgh|$$

$$= \ln|f| + \ln|g| + \ln|h|.$$

Using implicit differentiation, the derivative of the left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln|y|) = \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}.$$

Differentiating both sides and using this rule gives

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{f'}{f} + \frac{g'}{g} + \frac{h'}{h}.$$

Therefore, using logarithmic differentiation

$$y = f(x)g(x)h(x)$$

gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y\left(\frac{f'(x)}{f(x)} + \frac{g'(x)}{g(x)} + \frac{h'(x)}{h(x)}\right).$$

This could have also been arrived at by rearranging the product rule for three functions.

Example 2.3.2 (Lecture)**.** Use logarithmic differentiation to find the derivative of

$$y = \frac{\sqrt{(x+a)(x+b)}}{x-c}.$$

Solution

Take natural log of the absolute value of both sides:

$$\ln|y| = \ln\left|\frac{\sqrt{(x+a)(x+b)}}{x-c}\right|.$$

Use the product log rule:

$$\ln|y| = \ln\sqrt{x+a} + \ln\sqrt{x+b} - \ln|x-c|.$$

Solution (cont)

Use power log rule:

$$\ln|y| = \frac{1}{2}\ln|x+a| + \frac{1}{2}\ln|x+b| - \ln|x-c|.$$

Differentiate both sides with respect to x:

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2(x+a)} + \frac{1}{2(x+b)} - \frac{1}{x-c}.$$

Rearrange and substitute in expression for y:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sqrt{(x+a)(x+b)}}{x-c}\left(\frac{1}{2(x+a)} + \frac{1}{2(x+b)} - \frac{1}{x-c}\right).$$

Example 2.3.3 (Video)**.** Find the derivative, with respect to t, of

$$z = 3^{t^2-1}.$$

Solution

Take the natural logarithm (note $z > 0$) of both sides to give

$$\ln z = \ln\left|3^{t^2-1}\right| = (t^2-1)\ln(3).$$

Differentiate both sides with respect to t:

$$\frac{z'}{z} = 2t\ln 3.$$

Rearrange and substitute back in the expression for z to get

$$z' = 2t\ln 3\, z = 2t\ln 3\, 3^{t^2-1}.$$

Alternatively, you could rearrange to write as

$$z = e^{\ln(3)(t^2-1)}$$

and differentiate using the chain rule.

2.3.3 Parametric differentiation

A parametric function can also be differentiated. Consider the parametric function

$$y = f(t) \qquad x = g(t).$$

The derivative of y with respect to x is given by

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{f'(t)}{g'(t)} \tag{2.3.1}$$

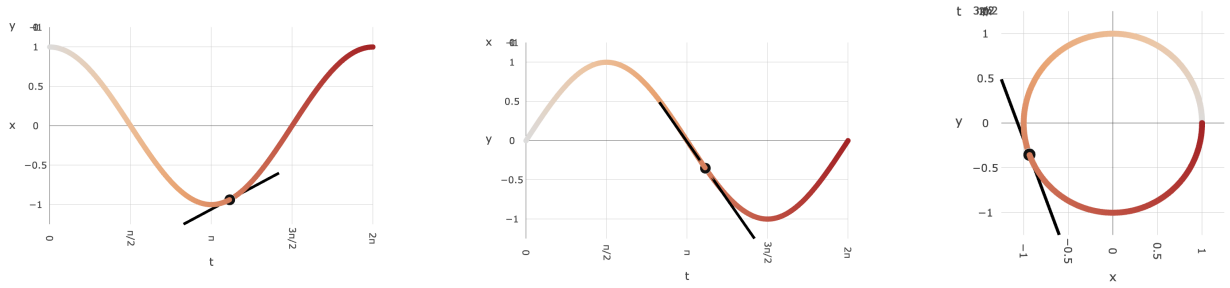
Consider, for example,

$$y = \sin t \qquad x = \cos t,$$

i.e. the unit circle. Here

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{\cos t}{-\sin t} = -\cot t.$$

The animation in figure 2.4 demonstrates this by adding the tangent line to the parametric form of the unit circle (as used in figure 1.8). Projecting to the x-t plane, the tangent line has gradient $\mathrm{d}x/\mathrm{d}t$, for the y-t plane it is $\mathrm{d}y/\mathrm{d}t$ and so as we project to the x-y plane the tangent has gradient $\frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}x}$. This is visualised by the relationship between the components of the tangent line in figure 2.5



(a) $x - t$ plane (b) $y - t$ plane (c) $x - y$ plane

Figure 2.4: Screenshots of the interactive plot of the parametric curve $x = \cos t$, $y = \sin t$, available in the html version of these notes. The tangent line in black shows the gradient at the highlighted point.

The process can be extended to the second derivative (and higher order derivatives). Define

$$z(t) = \frac{\mathrm{d}y}{\mathrm{d}x},$$

then repeat the process to find the derivative of z with respect to x:

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{dz}{\mathrm{d}x} = \frac{dz/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}.$$

Continuing the previous example,

$$y = \sin t \qquad x = \cos t \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\cot t.$$

Now

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{dz}{\mathrm{d}x} = \frac{dz/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}$$

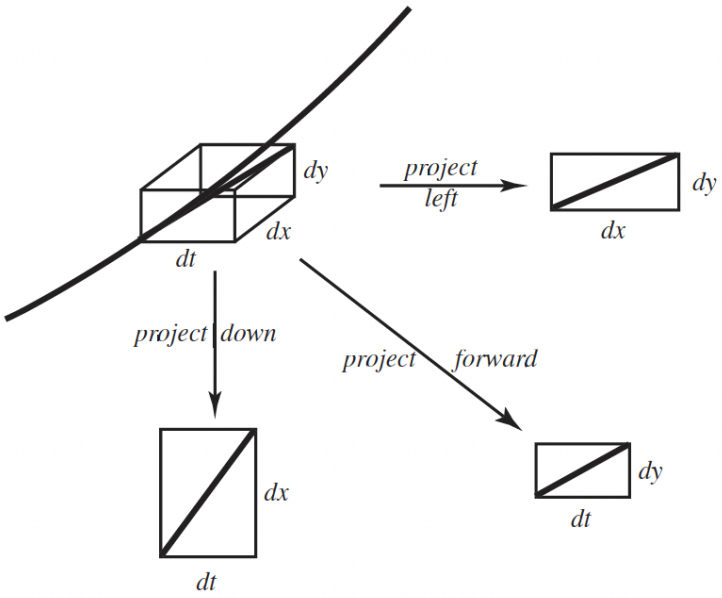


Figure 2.5: We can think of the tangent line as the ratio of the components of the derivatives $\mathrm{d}x$, $\mathrm{d}y$, and $\mathrm{d}t$, and so the relation $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}$ follows from simple geometry. (Figure from Tall 2013 "*How Humans Learn to Think Mathematically*", Chapter 11).

$$= \frac{d(-\cot t)/\mathrm{d}t}{-\sin t} = \frac{\csc^2 t}{-\sin t} = -\frac{1}{\sin^3 t}$$
$$= -\frac{1}{y^3}.$$

Note it is not always convenient to express the derivative in terms of x and y.

⋆ Exam mistake of the week ⋆

Using

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{d^2y/\mathrm{d}t^2}{d^2x/\mathrm{d}t^2}$$

instead of

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{dz/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}, \qquad \text{where} \qquad z = \frac{\mathrm{d}y}{\mathrm{d}x}.$$

25/26 Exam Q2(a)(i). The equations

$$x = \sin\left(\frac{\pi t}{6}\right) \qquad \text{and} \qquad y = t^2 - 3t + 3$$

describe a curve in parametric form. Find expressions for $\mathrm{d}y/\mathrm{d}x$ and $d^2y/\mathrm{d}x^2$ in terms of t.

Correct answer:

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{6\left[12\cos\left(\frac{\pi t}{6}\right) + (2t-3)\pi\sin\left(\frac{\pi t}{6}\right)\right]}{\pi^2\cos^3\left(\frac{\pi t}{6}\right)}. \qquad ✓$$

Using the incorrect definition

$$\frac{d^2y/\mathrm{d}t^2}{d^2x/\mathrm{d}t^2} = -\frac{72}{\pi^2\sin\left(\frac{\pi t}{6}\right)}. \qquad \text{X}$$

Example 2.3.4 (Lecture)**.** Find the derivative $\mathrm{d}y/\mathrm{d}x$ and the second derivative $d^2y/\mathrm{d}x^2$ given that

$$x = t^3 + t \qquad y = e^{2t}.$$

Solution

Find derivatives of x and y:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 3t^2 + 1 \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = 2e^{2t}.$$

Take the ratio:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{2e^{2t}}{3t^2+1}.$$

Use quotient rule for $dz/\mathrm{d}t$:

$$u(t) = 2e^{2t} \qquad \text{and} \qquad v(t) = 3t^2 + 1.$$

Find the derivatives:

$$u'(t) = 4e^{2t} \qquad \text{and} \qquad v'(t) = 6t.$$

Substitute in:

$$\frac{dz}{\mathrm{d}t} = \frac{u'v - uv'}{v^2} = \frac{4e^{2t}(3t^2+1) - 12te^{2t}}{(3t^2+1)^2}.$$

Solution (cont)

and therefore

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{dz/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{4(3t^2-3t+1)}{(3t^2+1)^3}e^{2t}.$$

Recap

Key points from section 2.3

- **Implicit differentiation** is performed by differentiating each term in an equation of the form $f(x,y)=0$ with respect to x and rearranging for $\mathrm{d}y/\mathrm{d}x$.
- **Logarithmic differentiation** of a function $y=g(x)$ involves taking the log of both sides first to simplify the RHS:
$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\ln(g(x)).$$
- To find $\mathrm{d}y/\mathrm{d}x$ when x and y are in **parametric form**, use the relationship
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}.$$

3 Integration

Intended Learning Outcomes

The relevant ILOs for this section are

- **[ILO4]** Recall and apply appropriate rules to integrate functions and use these to measure areas associated with curves or combinations of curves.

3.1 Revision of basic integration

Learning Outcomes

On completion of this subsection, students will be able to

- Calculate the definite and indefinite integral of basic mathematical functions

Differentiation and Integration. Integrating a function is the inverse of differentiation, and requires finding its "anti-derivative".

$$x^3 \underset{\text{Integrate}}{\overset{\text{Differentiate}}{\rightleftharpoons}} 3x^2.$$

The integration is denoted as

$$\int 3x^2 \, \mathrm{d}x = x^3 + C \,,$$

where $\int$ is the integral sign, $3x^2$ is the integrand, $\mathrm{d}x$ indicates integration with respect to x, and $x^3 + C$ is the integral. The integral is not unique, the constant C can take any value, since a constant differentiates to zero. This **constant of integration** must always be remembered when integrating.

Differentiation vs. Integration. Consider a function $y = f(x)$. The process of differentiation gives information about its rate of change. The process of integration gives information about the accumulation, $A(x)$, of the function, such that $A'(x) = f(x)$ (see figure 3.1).

Some standard integrals are given in table 2.

Using the chain rule for differentiation

$$\frac{\mathrm{d}}{\mathrm{d}x} f(ax + b) = a f'(ax + b) \,,$$

some of the integrals in table 2 can be generalised to the case of linear functions in the argument (see table 3).

Some other standard integrals to be aware of are given in table 4(and most are available in the mathematical formula tables). Note these can be derived using some of the techniques introduced later in

Function $y = f(x)$	**Integral** $\int f(x)\,\mathrm{d}x$
$x^n \ (n \neq -1)$	$\dfrac{1}{n+1}x^{n+1} + C$
$\dfrac{1}{x}$	$\ln\lvert x\rvert + C$
e^x	$e^x + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$

Table 2: Some standard integrals.

Function $y = f(x)$	**Integral** $\int f(x)\,\mathrm{d}x$
e^{ax+b}	$\dfrac{1}{a}e^{ax+b} + C$
$\sin(ax+b)$	$-\dfrac{1}{a}\cos(ax+b) + C$
$\cos(ax+b)$	$\dfrac{1}{a}\sin(ax+b) + C$

Table 3: Standard integrals extended to general linear functions, where a and b are constants.

this section.

Function $y = f(x)$	**Integral** $\int f(x)\,\mathrm{d}x$
$\tan x$	$\ln(\sec x) + C$
$\dfrac{1}{a^2+x^2}$	$\dfrac{1}{a}\tan^{-1}\left(\dfrac{x}{a}\right) + C$
$\dfrac{1}{\sqrt{a^2-x^2}}$	$\sin^{-1}\left(\dfrac{x}{a}\right) + C$
$\ln x \quad (x > 0)$	$x\ln x - x + C$

Table 4: Some more standard integrals that are given in the mathematical formula tables (which can be used in the exam).

Linearity of integration. The integral of a sum of functions is equal to the sum of integrals

$$\int [f(x) + g(x)]\,\mathrm{d}x = \int f(x)\,\mathrm{d}x + \int g(x)\,\mathrm{d}x.$$

Example 3.1.1 (Lecture)**.** Find the integral

$$\int (4\cos x - 3e^x)\ \mathrm{d}x.$$

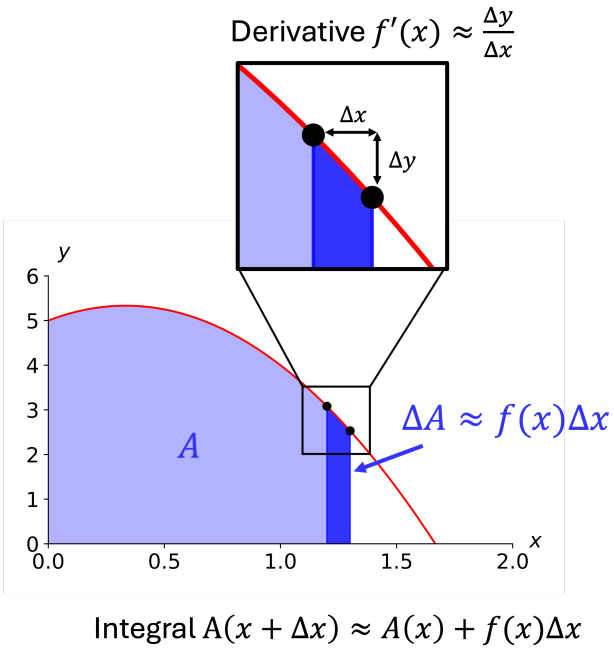


Figure 3.1: A sketch of the relationship between the derivative $f'(x)$ and the integral $f(x)$. If we zoom in, the curve is locally linear with gradient $\Delta y/\Delta x$ which is approximately equal to $f'(x)$. Integrating tells us how the area under the curve increases with x, so in moving from x to $x + \Delta x$ we accrue an extra area of $\Delta A \approx f(x)\Delta x$. The integral is the sum of these thin strips in the limit $\Delta x \to 0$, which depends on where you begin the sum from, hence the need for a constant of integration, such that $\int f'(x)\,\mathrm{d}x = f(x) + C$.

Solution

$$4\sin x - 3e^x + C.$$

The Definite Integral. The definite integral uses the notation

$$\int_a^b f(x)\,\mathrm{d}x.$$

It measures how much the function has accumulated (or, equivalently, the area under the curve) between $x = a$ and $x = b$. The values a and b are referred to as the 'limits of integration'. It is referred to as the "integral from a to b".

To calculate the definite integral, first calculate the indefinite integral or "anti-derivative".

$$F(x) = \int f(x)\,\mathrm{d}x.$$

Then, the definite integral is the difference between the indefinite integral at a and b

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Note, $F(x)$ contains a constant of integration, but this will always cancel out in the definite integral, so can be ignored.

It follows that the definite integral is additive, i.e. it can be 'split up'. If $a < b < c$,

$$\begin{aligned}\int_a^b f(x)\,\mathrm{d}x + \int_b^c f(x)\,\mathrm{d}x &= F(c) - F(b) + F(b) - F(a)\\ &= F(c) - F(a)\\ &= \int_a^c f(x)\,\mathrm{d}x.\end{aligned}$$

Example 3.1.2 (Video)**.** A metal strip has width w [cm] and is heated unevenly until it has a steady temperature field. The end of the strip is in contact with a conducting plate. The heat flux density at the interface is kh where

$$h = 4 + \left(\frac{x}{w}\right)^2 - 4\left(\frac{x}{w}\right)^3 \text{ [W/cm]}$$

where $0 < x < w$ [cm].
Calculate the total heat flux

$$H = k\int_0^w h\,\mathrm{d}x$$

in terms of the conductivity k and the width w.

Solution

Calculate the indefinite integral and substitute in the limits

$$\begin{aligned}\int_0^w h\,\mathrm{d}x &= \left[4x + \frac{x^3}{3w^2} - \frac{x^4}{w^3}\right]_0^w\\ &= 4w + \frac{w}{3} - w\\ &= 3w + \frac{w}{3} = \frac{10w}{3}.\end{aligned}$$

Then

$$H = k\int_0^w h\,\mathrm{d}x = \frac{10}{3}kw \text{ [W]}.$$

Recap

Key points from section 3.1

Recap (cont)

- Integration is the **reverse operation** to differentiation.
- Integration measures the **accumulation of a function**.
- Indefinite integrals give rise to a **constant of integration**.
- Definite integrals are calculated using the indefinite integral

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

3.2 Integration and areas defined by curves

Learning Outcomes

On completion of this subsection, students will be able to

- Calculate the area under a curve and between curves using integration

The definite integral provides a way of finding the area under or above a curve or the area between curves.

Area under a curve. Consider the curve in figure 3.2 defined by the function $y = f(x)$ for values of x between the values $x = a$ and $x = b$.

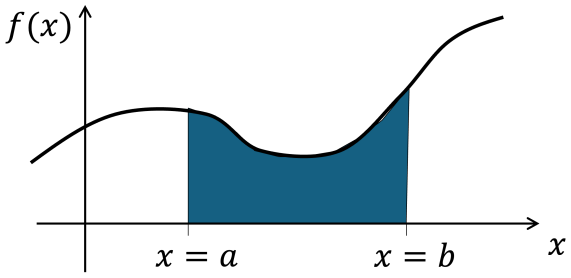


Figure 3.2: Sketch showing the area under the curve $f(x)$ given by $\int_a^b f(x)\,\mathrm{d}x$.

The area under the curve (*i.e.* between the curve and the x-axis) for $a \leq x \leq b$ is given by the integral

$$A = \int_a^b f(x)\,\mathrm{d}x.$$

Example 3.2.1 (DIY). Find the area under the curve

$$f(x) = 6x^2$$

between $2 \leq x \leq 5$.

Solution

The indefinite integral is

$$F(x) = \int f(x)\,\mathrm{d}x = \int 6x^2\,\mathrm{d}x$$
$$= 2x^3 + C.$$

Evaluating at the limits and subtracting

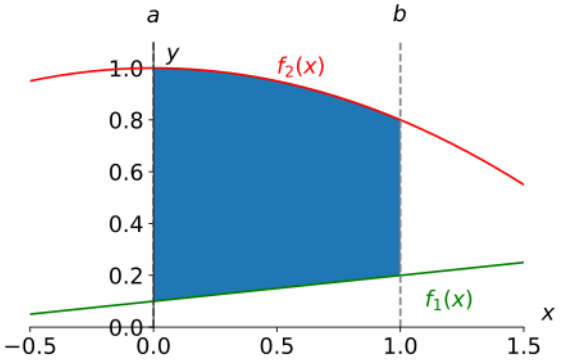
$$F(5) - F(2) = 2\left(5^3 - 2^3\right) = 2 \times 117$$
$$= 234.$$

Area between curves. The area under a curve is a special case of finding the area between two curves. The area is given by

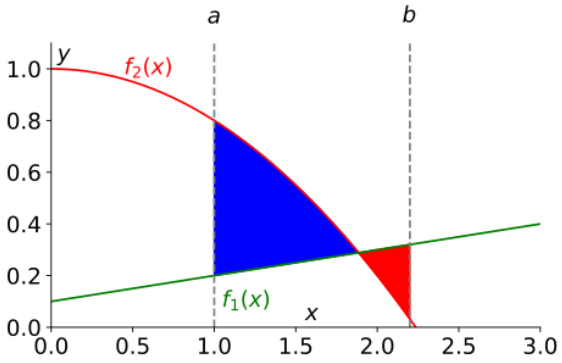
$$\int_a^b (f_2 - f_1)\,\mathrm{d}x$$

where f_1 is the lower curve and f_2 is the upper curve and a and b are the limits on the area (see figure 3.3(a)).

When using this formula, be aware of points where the curves cross. The area will be subtracted for regions where $f_2(x) < f_1(x)$ (see figure 3.3(a)).



(a) $f_2(x) > f_1(x)$ for $a \leq x \leq b$



(b) $f_2(x)$ and $f_1(x)$ cross in the domain $a \leq x \leq b$

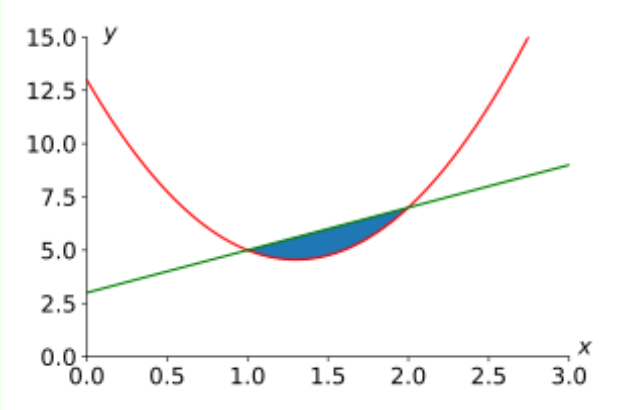
Figure 3.3: Visualisation of the area between curves $f_1(x)$ and $f_2(x)$ given by $\int_a^b [f_2(x) - f_1(x)]\,\mathrm{d}x$. Blue indicates a positive contribution to the integral, red a negative contribution. *I.e.* in (b) the result is the blue area minus the red area.

Example 3.2.2 (Lecture)**.** Find the area between the curves represented by

$$f(x) = 5x^2 - 13x + 13$$

and

$$g(x) = 2x + 3.$$



Solution

Find where $f(x)$ and $g(x)$ cross by solving $f(x) = g(x)$.

$$\begin{aligned} 5x^2 - 13x + 13 &= 2x + 3 \\ \therefore 5x^2 - 15x + 10 &= 0 \\ \therefore 5(x-1)(x-2) &= 0. \end{aligned}$$

Solving the quadratic gives $x = 1$ and $x = 2$.

Solution (cont)

Note that $g(x)$ passes above $f(x)$ and so the area between the curves is therefore

$$\begin{aligned}\int_1^2 [g(x) - f(x)]\,\mathrm{d}x &= \int_1^2 \left(-5x^2 + 15x - 10\right)\mathrm{d}x \\ &= \left[-\frac{5}{3}x^3 + \frac{15}{2}x^2 - 10x\right]_1^2 \\ &= \left[\frac{1}{6}(-80 + 180 - 120) - (-10 + 45 - 60)\right] \\ &= \frac{5}{6}.\end{aligned}$$

Recap

Key points from section 3.2

- The area **under the curve** for $a \leq x \leq b$ is

$$A = \int_a^b f(x)\,\mathrm{d}x.$$

- The area **between two curves** is

$$\int_a^b (f_2 - f_1)\,\mathrm{d}x.$$

- This assumes $f_2 > f_1$, so separate integrals are needed to deal with **crossing points**.

3.3 Improper integrals

Learning Outcomes

On completion of this subsection, students will be able to

- Identify improper integrals
- Use limits to investigate improper integrals

Consider the integral,

$$\int_0^1 x^2\,\mathrm{d}x.$$

- The integrand is finite
- The limits are finite

- The integrand can be integrated

This integral can be carried out without problems and so this is said to be a **proper integral**.

Integrals with infinite limits. An integral with an infinite limit, such as

$$\int_a^\infty f(x)\,\mathrm{d}x \qquad \text{or} \qquad \int_{-\infty}^b f(x)\,\mathrm{d}x \qquad \text{or} \qquad \int_{-\infty}^\infty f(x)\,\mathrm{d}x,$$

is said to be an **improper integral**. It **may** be possible to evaluate this integral or it may be infinite.

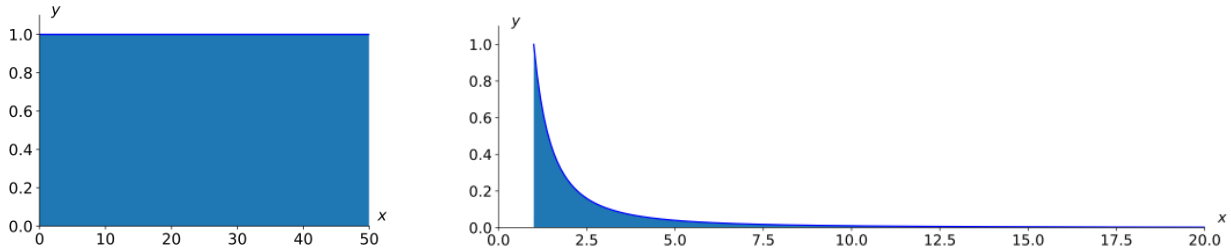
Sometimes a geometrical consideration can be insightful. Consider the integral

$$\int_1^\infty 1\,\mathrm{d}x.$$

From the area under the curve (figure 3.4(a)), it can be seen that the integral is infinite, *i.e.* this integral cannot be evaluated, its value is not defined. Instead, consider the integral

$$\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x.$$

The area under the curve is infinitely long but infinitely thin (figure 3.4(b)). It is not clear whether the integral will be finite or infinite.



(a) $\int_1^\infty 1\,\mathrm{d}x$ (b) $\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x$

Figure 3.4: Plots of the area under the curve for two example improper integrals.

Replacing the infinite limit with a parameter can usually be used to evaluate the integral. So, rather than $\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x$, consider the integral $\int_1^u \frac{1}{x^2}\,\mathrm{d}x$ and take the limit as u becomes very large:

$$\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = \lim_{u\to\infty} \int_1^u \frac{1}{x^2}\,\mathrm{d}x.$$

First, evaluate the substituted integral

$$\int_1^u \frac{1}{x^2}\,\mathrm{d}x = \left[-\frac{1}{x}\right]_1^u = -\frac{1}{u} - \left(-\frac{1}{1}\right) = 1 - \frac{1}{u}.$$

Then, take the limit

$$\lim_{u\to\infty} \int_1^u \frac{1}{x^2}\,\mathrm{d}x = \lim_{u\to\infty} \left(1 - \frac{1}{u}\right) = 1.$$

Therefore, in this case we see that the integral **converges** to a value of $\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = 1$.

On the other hand, consider $\int_1^\infty \frac{1}{x}\,\mathrm{d}x$. Making the substitution

$$\int_1^u \frac{1}{x}\,\mathrm{d}x = [\ln x]_1^u = \ln u\,,$$

and so

$$\int_1^\infty \frac{1}{x}\,\mathrm{d}x = \lim_{u\to\infty} \ln u$$

which becomes infinite, so this integral **does not converge**.

Example 3.3.1 (Lecture)**.** Find out whether the following improper integrals converge

(a) $\displaystyle\int_1^\infty \frac{1}{x^3}\,\mathrm{d}x$

(b) $\displaystyle\int_1^\infty \frac{x}{1+x^2}\,\mathrm{d}x$

and if so, to what value.

Hint:

$$\int \frac{x}{1+x^2}\,\mathrm{d}x = \frac{1}{2}\ln(1+x^2) + C.$$

Solution

(a) Substitute the upper limit of the integral and take the limit:

$$\int_1^\infty \frac{1}{x^3}\,\mathrm{d}x = \lim_{u\to\infty}\int_1^u x^{-3}\,\mathrm{d}x = \lim_{u\to\infty}\left[-\frac{1}{2x^2}\right]_1^u$$

$$= \lim_{u\to\infty}\left(\frac{1}{2} - \frac{1}{2u^2}\right) = \frac{1}{2}.$$

(b) Substitute the upper limit of the integral and calculate the integral:

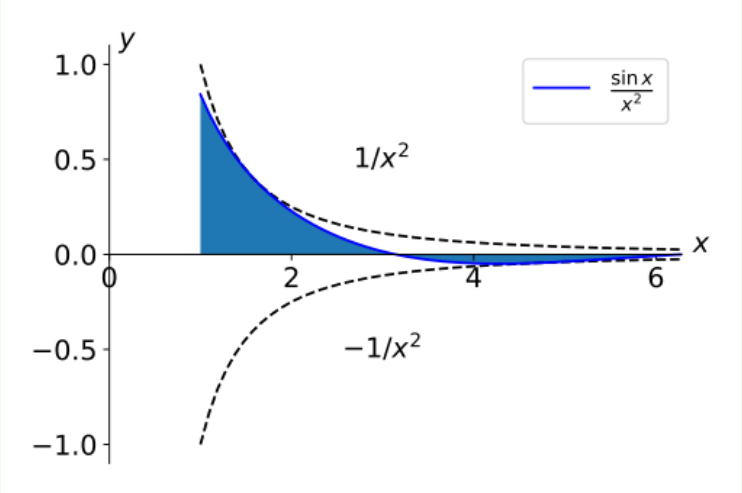
$$\int_1^\infty \frac{x}{1+x^2}\,\mathrm{d}x = \lim_{u\to\infty}\int_1^u \frac{x}{1+x^2}\,\mathrm{d}x = \lim_{u\to\infty}\left[\frac{1}{2}\ln(1+x^2)\right]_1^u$$

$$= \lim_{u\to\infty}\frac{1}{2}\ln(1+u^2) - \frac{1}{2}\ln 2 = \infty.$$

Sometimes, even when an integral cannot be calculated, it is possible to put some bounds on it by using an integral we can calculate.

Example 3.3.2 (DIY). Find lower and upper bounds on the integral

$$\int_1^\infty \frac{\sin x}{x^2}\,\mathrm{d}x.$$



Solution

Since

$$-1 \le \sin x \le 1,$$

then

$$-\frac{1}{x^2} \le \frac{\sin x}{x^2} \le \frac{1}{x^2}.$$

We already showed $\int_1^\infty 1/x^2 = 1$, therefore

$$-1 \le \int_1^\infty \frac{\sin x}{x^2}\,\mathrm{d}x \le 1.$$

Improper Integrals with a singularity in the integrand. Consider now the integral

$$\int_{-1}^{1} \frac{1}{x(1-x^2)}\,\mathrm{d}x\,.$$

Although the limits are finite, the integrand becomes infinite

- at either limit
- at a point between the limits

as shown in figure 3.5. When either of these are true, it is an **improper integral**.

To evaluate such an integral, once again the concept of a limit can be used. Consider the simpler

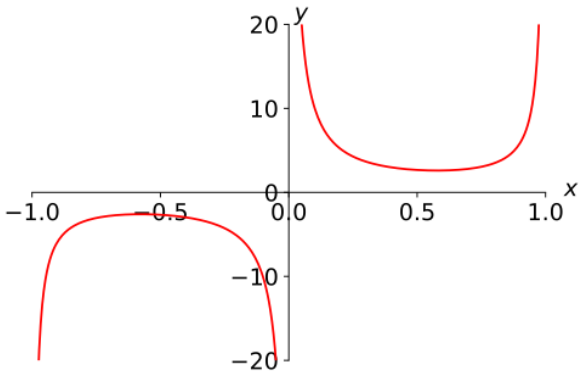


Figure 3.5: Plot of the function $\frac{1}{x(1-x^2)}$ which goes to infinity at $x = -1$, 0, and 1.

integral $\int_0^1 \frac{1}{x}\,\mathrm{d}x$ where the integrand is infinite at $x = 0$. Rewrite as

$$\int_0^1 \frac{1}{x}\,\mathrm{d}x = \lim_{u\to 0}\int_u^1 \frac{1}{x}\,\mathrm{d}x = \lim_{u\to 0}\left[\ln x\right]_u^1 = \lim_{u\to 0}(-\ln u),$$

which does not converge as $\ln(0)$ does not exist.

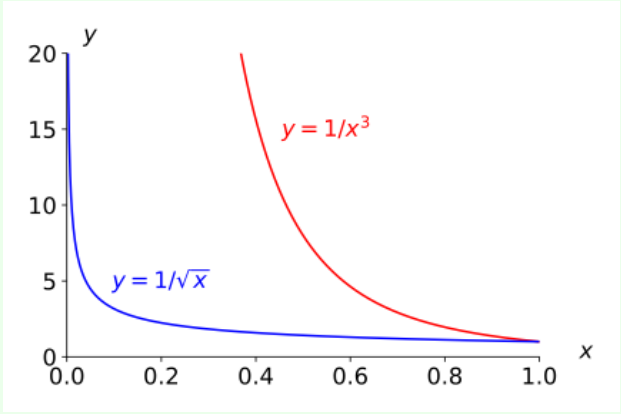
Example 3.3.3 (Video)**.** Investigate the improper integrals

$$\int_0^1 \frac{1}{x^3}\,\mathrm{d}x$$

and

$$\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x$$

to determine whether they converge, and if so, to what value.



Solution

Both integrands diverge as $x \to 0$, so we evaluate in the limit. First case:

$$\int_0^1 \frac{1}{x^3}\,\mathrm{d}x = \lim_{u\to 0}\int_u^1 x^{-3}\,\mathrm{d}x = \lim_{u\to 0}\left[-\frac{x^{-2}}{2}\right]_u^1$$

$$= \lim_{u\to 0}\left(\frac{1}{2u^2} - \frac{1}{2}\right) = \text{undefined}.$$

In the second case

$$\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim_{u\to 0}\int_u^1 x^{-1/2}\,\mathrm{d}x = \lim_{u\to 0}\left[2x^{1/2}\right]_u^1$$

$$= \lim_{u\to 0}\left(2 - 2u^{1/2}\right) = 2.$$

Convergence of x^n. For which values of n do the following integrals converge?

(a) $\displaystyle\int_0^1 x^n\,\mathrm{d}x$

(b) $\displaystyle\int_1^\infty x^n\,\mathrm{d}x$

(c) $\displaystyle\int_0^\infty x^n\,\mathrm{d}x.$

First, let's calculate the indefinite integral, remembering the special case when $n = -1$

$$\int x^n\,\mathrm{d}x = \begin{cases} \dfrac{x^{n+1}}{n+1} + C, & n \neq -1, \\ \ln|x| + C, & n = -1. \end{cases}$$

(a) The integrand has a singularity at $x = 0$ if $n < 0$. From the indefinite integral

$$\int_0^1 x^n\,\mathrm{d}x = \begin{cases} \dfrac{1}{n+1} - \lim_{x\to 0}\dfrac{x^{n+1}}{n+1}, & n \neq -1, \\ -\lim_{x\to 0}\ln|x|, & n = -1. \end{cases}$$

Inspecting the limits $\lim_{x\to 0}\frac{x^{n+1}}{n+1}$, converges if $n > -1$ and does not converge otherwise. Additionally, $\lim_{x\to 0}\ln|x|$ does not converge. So, $\int_0^1 x^n\,\mathrm{d}x$ **converges for** $n > -1$. See figure 3.6(a) for some example curves.

(b) The integrand goes to infinity as $x \to \infty$ if $n > 0$. From the indefinite integral

$$\int_1^\infty x^n \, \mathrm{d}x = \begin{cases} \lim_{x\to\infty} \dfrac{x^{n+1}}{n+1} - \dfrac{1}{n+1}, & n \neq -1, \\ \lim_{x\to\infty} \ln|x|, & n = -1. \end{cases}$$

Inspecting the limits $\lim_{x\to\infty} \frac{x^{n+1}}{n+1}$, converges if $n < -1$ and does converge not otherwise. Additionally, $\lim_{x\to\infty} \ln|x|$ does not converge. So, $\int_1^\infty x^n \, \mathrm{d}x$ **converges for** $n < -1$. See figure 3.6(b) for some example curves.

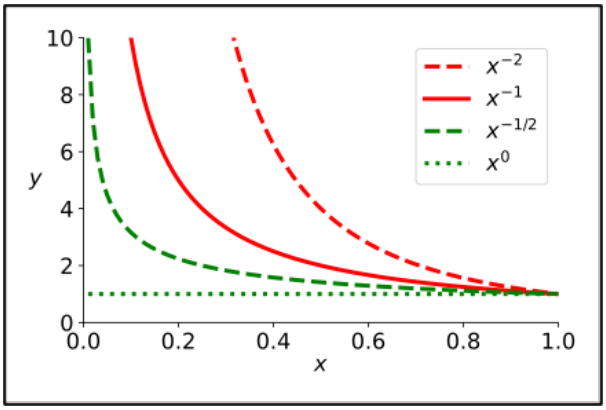
(c) The integrand goes to infinity as $x \to 0$ if $n < 0$ and as $x \to \infty$ if $n > 0$. We can split the limits to write the integral as the sum of the integrals in (a) and (b)

$$\int_0^\infty x^n \, \mathrm{d}x = \int_0^1 x^n \, \mathrm{d}x + \int_1^\infty x^n \, \mathrm{d}x.$$

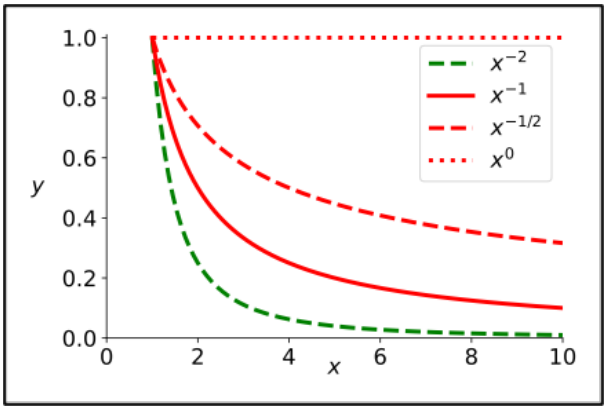
The first integral only converges for $n > -1$. The second integral only converges for $n < -1$, therefore

$$\int_0^\infty x^n \, \mathrm{d}x$$

does not converge for any n.



(a) $\int_0^1 x^n \, \mathrm{d}x$



(b) $\int_1^\infty x^n \, \mathrm{d}x$

Figure 3.6: Plots of various functions of the form x^n between the limits (a) $0 \leq x \leq 1$ and (b) $x \geq 1$. The green curves show those which result in a convergent improper integral in the limits labelled, while the integrals of the red curves do not converge.

Singularity between the limits of integration. If there is a place between the limits where the integrand is infinite, then the integral should be split at that point and the parts evaluated separately. If any of the parts does not converge then the whole integral does not converge.

⋆ Exam mistake of the week ⋆

Assuming that the integral cannot be evaluated if the integrand is infinite.

25/26 Exam Q3(a)(ii)

$$f(x) = 3x + 1 + \frac{2}{x-3} - \frac{4}{x+2}.$$

Determine whether

$$\int_4^6 f(x)\,dx$$

and

$$\int_0^3 f(x)\,dx$$

converge to a value and, if so, find the value.

The second integral does not converge, but the reason is not "because $\frac{2}{x-3}$ is undefined at $x = 3$" it is because $2\ln|x-3|$ is undefined at $x = 3$.

In this case this still resulted in the correct response, but would be incorrect if the term was *e.g.* $\frac{2}{\sqrt{x-3}}$.

Example 3.3.4 (Lecture)**.** Consider the improper integral

$$\int_{-1}^{1} \frac{1}{x^2}\,\mathrm{d}x.$$

(a) Determine whether it converges. If so, find its value.

(b) Compare this to the result you would find if you did not split the integral.

Solution

(a) Calculate the indefinite integral

$$\int x^{-2}\,\mathrm{d}x = -\frac{1}{x} + C.$$

Split the integral at the point where the integrand diverges:

$$\int_{-1}^{1} \frac{1}{x^2}\,\mathrm{d}x = \int_{-1}^{0} \frac{1}{x^2}\,\mathrm{d}x + \int_{0}^{1} \frac{1}{x^2}\,\mathrm{d}x.$$

Evaluate the integrals in the limit where the upper/lower limit of integration goes to 0:

$$\lim_{u\to 0^-} \left[-\frac{1}{x}\right]_{-1}^{u} = \infty,$$

$$\lim_{u\to 0^+} \left[-\frac{1}{x}\right]_{u}^{1} = \infty.$$

Solution (cont)

Therefore, $\int_{-1}^{1} \frac{1}{x^2}\,\mathrm{d}x$ does not converge.

(b) Naïvely substituting in the limits of integration gives

$$\left[-\frac{1}{x}\right]_{-1}^{1} = -2,$$

which is misleading (and wrong!) because the integral must be split at the singularity.

Recap

Key points from section 3.3

- A definite integral is improper if
  - Either or both **limits are infinite**
  - The **integrand becomes infinite** between or at the limits
- Improper integrals **may converge to a solution**, or may not.
- To investigate an improper integral, **substitute** the problem limit(s) for a variable **then take the limit** of that variable going to the original.

3.4 Further rules of integration

Learning Outcomes

On completion of this subsection, students will be able to

- Use integration by substitution to evaluate appropriate integrals
- Recall and apply the formula for integration by parts
- Use partial fractions to evaluate integrals involving rational fractions

3.4.1 Integration by substitution

In some cases, substituting the variable in an integral appropriately can make it easier to evaluate. In a sense, integration by substitution is like differentiating using the chain rule, but in reverse.

For example, consider the derivative of $\sin(x^2)$:

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x^2) = \cos(x^2) \times \frac{\mathrm{d}}{\mathrm{d}x}x^2 = 2x\cos(x^2).$$

So,

$$\int 2x\cos(x^2)\,\mathrm{d}x = \sin(x^2) + C.$$

But how to find this result without being given the answer to start with?

Start with the integral

$$\int 2x\cos(x^2)\,\mathrm{d}x.$$

We can see that the cosine is acting on x^2, so this is the candidate for our "inner function". So, substitute x^2 for a new variable, call it u, and rearrange the integral in terms of u.

$$u = x^2 \quad \text{and so} \quad \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \qquad \therefore\ \mathrm{d}u = 2x\,\mathrm{d}x.$$

Therefore, the integral becomes

$$\begin{aligned}\int \cos u\,\mathrm{d}u &= \sin u + C \\ &= \sin(x^2) + C.\end{aligned}$$

More generally, for an integral of the form

$$\int f(g(x))g'(x)\,\mathrm{d}x,$$

a substitution $u = g(x)$ can be used, resulting in the integral $\int f(u)\,\mathrm{d}u$. If this integral can be evaluated for u, then $u = g(x)$ can be substituted back into the result to give the solution in terms of x.

This can also be derived directly from the chain rule. Starting with a function of the form $y = F(g(x))$, the chain rule gives $\frac{\mathrm{d}y}{\mathrm{d}x} = g'(x)F'(g(x))$. Therefore,

$$\int f[g(x)]g'(x)\,\mathrm{d}x = F(g(x)) + C,$$

where $f(u) = F'(u)$.

Example 3.4.1 (DIY)**.** Using a substitution, evaluate the integral

$$\int \frac{(\ln x)^5}{x}\,\mathrm{d}x.$$

Solution

Define $u = \ln(x)$, and so $\mathrm{d}u = \dfrac{1}{x}\,\mathrm{d}x$. Substituting this in gives

$$\int \frac{(\ln x)^5}{x}\,\mathrm{d}x = \int u^5\,\mathrm{d}u = \frac{u^6}{6} + C.$$

Substituting $u = \ln(x)$ back in gives

$$\int \frac{(\ln x)^5}{x}\,\mathrm{d}x = \frac{1}{6}(\ln x)^6 + C.$$

Definite Integration by substitution. To evaluate a definite integral by substitution, the limits must be expressed in terms of the new variable. For example,

$$\int_0^2 x^2(x^3+5)^4\,\mathrm{d}x,$$

use the substitution $u(x) = x^3 + 5$ so that $\mathrm{d}u = 3x^2\,\mathrm{d}x$ and the integral becomes $\int \frac{1}{3}u^4\,\mathrm{d}u$. The limits need to be converted to u values. $u(0) = 5$ and $u(2) = 13$, so the integral becomes

$$\int_5^{13} \frac{1}{3}u^4\,\mathrm{d}u = \frac{1}{15}\left[u^5\right]_5^{13} = \frac{13^5 - 5^5}{15} = \frac{368168}{15}.$$

Example 3.4.2 (Lecture)**.** Using substitution, evaluate the integral

$$\int_0^{1/2} x\cos(\pi x^2)\,\mathrm{d}x.$$

Solution

Substitute $u = \pi x^2$, then $u' = 2\pi x$.

The limits of the integral become $u(0) = 0$ and $u(1/2) = \pi/4$.

As $\mathrm{d}u = u'\,\mathrm{d}x$, then $\mathrm{d}x = \dfrac{1}{2\pi x}\,\mathrm{d}u$, substituting gives

$$\begin{aligned}\int_0^{1/2} x\cos(\pi x^2)\,\mathrm{d}x &= \int_0^{\pi/4} \frac{x}{2\pi x}\cos u\,\mathrm{d}u \\ &= \frac{1}{2\pi}\int_0^{\pi/4} \cos u\,\mathrm{d}u\,.\end{aligned}$$

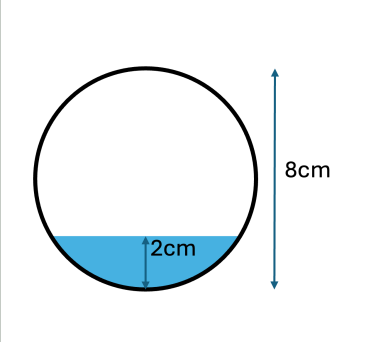
Solution (cont)

Evaluating the integral in u we get

$$\frac{1}{2\pi}\int_0^{\pi/4}\cos u\,\mathrm{d}u = \frac{1}{2\pi}\left[\sin u\right]_0^{\pi/4} = \frac{1}{2\pi}\times\frac{\sqrt{2}}{2}$$

$$= \frac{\sqrt{2}}{4\pi}.$$

Example 3.4.3 (Video)**.** A cylindrical tube of diameter 8cm and length 1m, held horizontally, is partially filled with water to a depth of 2cm.
Use integration to calculate the volume of water in the tube.



Solution

The cross section of the tube is a circle with radius $R = 4$cm, so can be represented by the equation:

$$x^2 + y^2 = 16$$

where the circle centre is at the origin $(0, 0)$.

The surface of the water lies on the straight horizontal line

$$y = -2.$$

We need to compute the integral below this curve and above the lower half of the circle, which is given by

$$y = -\sqrt{16 - x^2}.$$

To find the limits of integration, find where the two curves cross by substituting the equation for the line into the circle

$$x^2 + (-2)^2 = 16 \qquad \therefore\ x^2 = 12.$$

Solution (cont)

$$x = \pm 2\sqrt{3}.$$

Integrating the horizontal line gives the (negative) of the area above it

$$A_1 = \int_{-2\sqrt{3}}^{2\sqrt{3}} -2 \, \mathrm{d}x = -8\sqrt{3}.$$

Next, we need to integrate the lower semi-circle

$$A_2 = -\int_{-2\sqrt{3}}^{2\sqrt{3}} \sqrt{16 - x^2} \, \mathrm{d}x.$$

Using the substitution $x = 4\sin\left(\frac{t}{2}\right)$ then $\mathrm{d}x = 2\cos\left(\frac{t}{2}\right) \, \mathrm{d}t$.

The limits of the integral are at

$$t = 2\sin^{-1}\left(\pm\frac{\sqrt{3}}{2}\right) = \pm\frac{2\pi}{3}.$$

$$A_2 = -\int_{-2\pi/3}^{2\pi/3} 4\sqrt{1 - \sin^2\left(\frac{t}{2}\right)} \times 2\cos\left(\frac{t}{2}\right) \, dt = -8\int_{-2\pi/3}^{2\pi/3} \cos^2\left(\frac{t}{2}\right) \, dt.$$

Using the trigonometric identity

$$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

this becomes

$$A_2 = -4\int_{-2\pi/3}^{2\pi/3} (1 + \cos t) \, dt = -4\left[t + \sin t\right]_{-2\pi/3}^{2\pi/3} = -\frac{16\pi}{3} - 8\sin\left(\frac{2\pi}{3}\right).$$

Therefore, the area of the cross section filled by water is

$$A_1 - A_2 = -8\sqrt{3} + \frac{16\pi}{3} + 8\sin\left(\frac{2\pi}{3}\right) \approx 9.83 \text{ cm}^2.$$

So the volume of water is approximately

$$100 \times 9.83 = 983 \text{ cm}^3 \text{ or } 983\text{ml}.$$

3.4.2 Integration by parts

Often, a product of two functions can be integrated using a process known as 'integration by parts'. In a sense, integrating by parts is the reverse of differentiating using the product rule.

The basic formula for integration by parts can be derived from the product rule. Start with the function

$$y = u(x)v(x) \qquad \text{therefore} \qquad y'(x) = u'(x)v(x) + u(x)v'(x).$$

Taking the integral of both sides gives $\int y'(x)\,\mathrm{d}x = \int u'(x)v(x)\,\mathrm{d}x + \int u(x)v'(x)\,\mathrm{d}x$. Noting that $\int y'(x)\,\mathrm{d}x = y = u(x)v(x)$, we arrive at the formula for integration by parts

$$\int u(x)v'(x)\,\mathrm{d}x = u(x)v(x) - \int u'(x)v(x)\,\mathrm{d}x. \tag{3.4.1}$$

It can be used to integrate a product by defining one factor as $u(x)$ and the other as $v'(x)$, with the aim of making the resulting integral easier.

Take the integral $\int x\sin x\,\mathrm{d}x$, let $u(x) = x$ and $v'(x) = \sin x$. Thus, $u'(x) = 1$ and $v(x) = -\cos x$. Using equation (3.4.1)

$$\begin{aligned}\int x\sin x\,\mathrm{d}x &= -x\cos x + \int \cos x\,\mathrm{d}x.\\ &= -x\cos x + \sin x + C.\end{aligned}$$

Sometimes, integration by parts needs to be used more than once. For example, the function $x^2\cos x$ integrates (using equation (3.4.1) with $u = x^2$ and $v = \cos x$) to

$$\begin{aligned}\int x^2\cos x\,\mathrm{d}x &= x^2\sin x - \int 2x\sin x\,\mathrm{d}x\\ &= x^2\sin x - \left(-2x\cos x + \int 2\cos x\,\mathrm{d}x\right)\\ &= (x^2-2)\sin x + 2x\cos x + C.\end{aligned}$$

Example 3.4.4 (Lecture). Using integration by parts, find

$$\int e^x\sin x\,\mathrm{d}x.$$

Solution

Define $u = \sin x$ and $v' = e^x$. Then $u' = \cos(x)$ and $v = e^x$.

Using equation (3.4.1) $\int uv'\,\mathrm{d}x = uv - \int u'v\,\mathrm{d}x$ so,

$$\int e^x\sin x\,\mathrm{d}x = e^x\sin x - \int e^x\cos x\,\mathrm{d}x. \tag{1}$$

Solution (cont)

Now define $u = \cos(x)$ and $v' = e^x$. Then $u' = -\sin(x)$ and $v = e^x$.

$$\int e^x \cos x \, \mathrm{d}x = e^x \cos x + \int e^x \sin x \, \mathrm{d}x. \tag{2}$$

Define $I = \int e^x \sin x \, \mathrm{d}x$ then we can write equation (2) as

$$\int e^x \cos x \, \mathrm{d}x = e^x \cos x + I.$$

Substituting into equation (1),

$$I = e^x \sin x - e^x \cos x - I \qquad \therefore \; 2I = e^x(\sin x - \cos x)\,.$$

And so

$$\int e^x \sin x \, \mathrm{d}x = \frac{e^x}{2}(\sin x - \cos x) + C.$$

Usually, if the integrand is a product with a polynomial, then the polynomial is taken to be $u(x)$ so that $u'(x)$ will present a simpler integral. An exception is if the other factor is a logarithm or an inverse trigonometric function.

For example, to find $\int x^2 \ln x \, \mathrm{d}x$, define $u(x) = \ln x$ and $v'(x) = x^2$, so that $u'(x) = \frac{1}{x}$ and $v(x) = \frac{x^3}{3}$. Using equation (3.4.1) the integral becomes

$$\begin{aligned}\int x^2 \ln x \, \mathrm{d}x &= \frac{1}{3}x^3 \ln x - \int \frac{x^3}{3x} \, \mathrm{d}x \\ &= \frac{1}{3}x^3 \ln x - \frac{1}{3}\int x^2 \, \mathrm{d}x \\ &= \frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C.\end{aligned}$$

Example 3.4.5 (Lecture)**.** Using integration by parts, find

$$\int \tan^{-1} x \, \mathrm{d}x.$$

Solution

Define $u = \tan^{-1} x$ and $v' = 1$. Then $u' = \dfrac{1}{1+x^2}$ and $v = x$.

Using the formula for IBP (equation (3.4.1))

$$\int \tan^{-1} x \, \mathrm{d}x = x \tan^{-1} x - \int \frac{x}{1+x^2} \, \mathrm{d}x.$$

Solution (cont)

Using the substitution $z = x^2$, and so $dz = 2x\,\mathrm{d}x$,

$$\begin{aligned}\int \frac{x}{1+x^2}\,\mathrm{d}x &= \frac{1}{2}\int \frac{1}{1+z}\,\mathrm{d}z \\ &= \frac{1}{2}\ln(1+z) = \frac{1}{2}\ln(1+x^2).\end{aligned}$$

Therefore,

$$\int \tan^{-1} x\,\mathrm{d}x = x\tan^{-1} x - \frac{1}{2}\ln(1+x^2) + C.$$

$$\int u(x)v'(x)\,\mathrm{d}x = u(x)v(x) - \int u'(x)v(x)\,\mathrm{d}x.$$

Note that, to carry out definite integration using integration by parts, all terms need to be evaluated at the limits, so

$$\int_a^b u(x)v'(x)\,\mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,\mathrm{d}x.$$

3.4.3 Integration by partial fractions

If a function can be expressed in terms of partial fractions, then these partial fractions can generally be integrated.

Example 3.4.6 (Video)**.** Find

$$\int \frac{x+17}{(x+3)(2x-1)}\,\mathrm{d}x$$

using integration by partial fractions.

Solution

First write the partial fractions form

$$\frac{x+17}{(x+3)(2x-1)} = \frac{A}{x+3} + \frac{B}{2x-1}\,.$$

Multiply through by the denominator

$$x + 17 = A(2x-1) + B(x+3)\,.$$

Setting $x = -3$ allows us to find A:

$$14 = -7A \qquad \therefore \qquad A = -2.$$

Solution (cont)

Setting $x = 1/2$ allows us to find B:

$$\frac{35}{2} = \frac{7}{2}B \qquad \therefore \qquad B = 5.$$

Now evaluate the integral

$$\begin{aligned}\int \frac{x+17}{(x+3)(2x-1)}\,\mathrm{d}x &= -2\int \frac{1}{x+3}\,\mathrm{d}x + 5\int \frac{1}{2x-1}\,\mathrm{d}x \\ &= -2\ln|x+3| + \frac{5}{2}\ln|2x-1| + C.\end{aligned}$$

Recap

Key points from section 3.4

- **Integration by substitution** is useful for integrands that can be expressed in the form

$$\int f(g(x))g'(x)\,\mathrm{d}x.$$

- **Integration by parts** is used for integrating products of functions

$$\int u(x)v'(x)\,\mathrm{d}x = u(x)v(x) - \int u'(x)v(x)\,\mathrm{d}x.$$

- **Partial fractions** can be used to make suitable integrals easier.

3.5 Double integrals over rectangular regions

Learning Outcomes

On completion of this subsection, students will be able to

- Calculate the volume under a surface with rectangular bounds using integration

The integral $\int_a^b f(x)\,\mathrm{d}x$ represents the area under the curve $y = f(x)$ between $x = a$ and $x = b$. Consider instead a function of two variables $f(x, y)$. The integral

$$\int_c^d f(x, y)\,\mathrm{d}x$$

represents the area of a slice at position y. Multiplying by $\mathrm{d}y$ gives $\left(\int_c^d f(x, y)\,\mathrm{d}x\right)\,\mathrm{d}y$ representing the volume of a slice at position y with infinitesimal thickness $\mathrm{d}y$. The sum of all slices between $y = a$ and

$y = b$ is given by the integral of the integral, *i.e.*

$$\int_a^b \int_c^d f(x, y)\, \mathrm{d}x\, \mathrm{d}y\,. \tag{3.5.1}$$

The double integral in equation (3.5.1) represents the volume underneath the function but above the rectangle $z = 0$; $a \leq y \leq b$; $c \leq x \leq d$.

Figure 3.7 illustrates the double integration process.

Inner and outer integral. From the definition in equation (3.5.1) the integral $\int_c^d f(x, y)\, \mathrm{d}x$ is known as the 'inner integral'. It can be evaluated and will be a function of y

$$\int_c^d f(x, y)\, \mathrm{d}x = g(y).$$

To find $g(y)$, treat y as a constant and integrate with respect to x. *E.g.* If $f(x, y) = e^x \cos y$, then

$$\int_c^d f(x, y)\, \mathrm{d}x = \cos y \left[e^d - e^c\right].$$

Once the inner integral is evaluated it can be placed in the main integral to give

$$\int_a^b g(y)\, \mathrm{d}y.$$

This is known as the 'outer integral'.

Note that reversing the order of the integrations (along with their limits) preserves the integral, *i.e.*

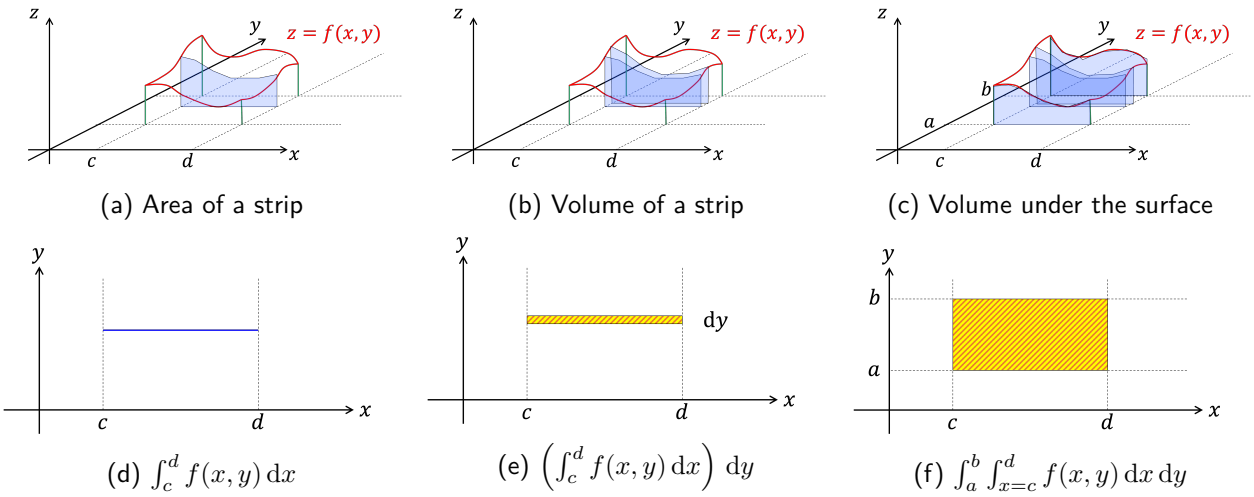
$$\int_a^b \int_{x=c}^d f(x, y)\, \mathrm{d}x\, \mathrm{d}y = \int_c^d \int_{y=a}^b f(x, y)\, \mathrm{d}y\, \mathrm{d}x.$$

Example 3.5.1 (DIY)**.** Find the double integral

$$\int_{-\pi}^{\pi} \int_0^1 (x^2 + 2x - 1) \cos\left(\frac{y}{2}\right)\, \mathrm{d}x\, \mathrm{d}y$$

and show it is the same if the order of integration is reversed

$$\int_0^1 \int_{-\pi}^{\pi} (x^2 + 2x - 1) \cos\left(\frac{y}{2}\right)\, \mathrm{d}y\, \mathrm{d}x.$$



(a) Area of a strip (b) Volume of a strip (c) Volume under the surface

(d) $\int_c^d f(x,y)\,\mathrm{d}x$ (e) $\left(\int_c^d f(x,y)\,\mathrm{d}x\right)\,\mathrm{d}y$ (f) $\int_a^b \int_{x=c}^d f(x,y)\,\mathrm{d}x\,\mathrm{d}y$

Figure 3.7: Building up the double integral to find the volume under a surface step-by-step from (a)–(c) three-dimensional view and (d)–(f) looking down from above.

Solution

Take the first case and evaluate the inner integral, treating y as constant:

$$\int_0^1 (x^2+2x-1)\cos\left(\frac{y}{2}\right)\,\mathrm{d}x = \cos\left(\frac{y}{2}\right)\left[\frac{x^3}{3}+x^2-x\right]_0^1 = \frac{1}{3}\cos\left(\frac{y}{2}\right).$$

Now evaluate the outer integral:

$$\frac{1}{3}\int_{-\pi}^{\pi}\cos\left(\frac{y}{2}\right)\,\mathrm{d}y = \frac{2}{3}\left[\sin\left(\frac{y}{2}\right)\right]_{-\pi}^{\pi} = \frac{4}{3}.$$

Reversing the order and evaluating the inner integral first:

$$\int_{-\pi}^{\pi}(x^2+2x-1)\cos\left(\frac{y}{2}\right)\,\mathrm{d}y = (x^2+2x-1)\left[2\sin\left(\frac{y}{2}\right)\right]_{-\pi}^{\pi}$$

$$= 4(x^2+2x-1).$$

Now evaluate the outer integral:

$$4\int_0^1 (x^2+2x-1)\,\mathrm{d}x = 4\left[\frac{x^3}{3}+x^2-x\right]_0^1 = \frac{4}{3}.$$

Example 3.5.2 (Video)**.** An experiment is set up to measure the diffusion coefficient of benzene in water.
The flux at the outlet F is measured, and should be given approximately by

$$\frac{Fh}{D} = \int_0^l \int_0^l c(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

where D is the diffusion constant and $c(x,y)$ the concentration at the inlet.
To correctly estimate D, the experiment is meant to keep $c(x,y) = c_0$ (a constant). However, due to an error in the setup, the actual concentration is

$$c(x,y) = \left(\frac{3}{4}x^2 + y + xy\right) c_0.$$

Given that F is measured to be 2.5 [mol/s], $h = 0.04$ [cm], $l = 1$ [cm] and $c_0 = 10^4$ [mol/cm^3] find

- the real value of D (using the actual concentration $c = \left(\frac{3}{4}x^2 + y + xy\right) c_0$)
- the value of D obtained by the researchers (assuming $c = c_0$)

Solution

First take the case where

$$c(x,y) = \left(\frac{3}{4}x^2 + y + xy\right) c_0$$

and compute the inner integral

$$I(y) = \int_0^1 c(x,y)\,\mathrm{d}x = c_0 \left[\frac{1}{4}x^3 + yx + \frac{1}{2}x^2 y\right]_0^1$$

$$= c_0 \left(\frac{1}{4} + \frac{3}{2}y\right).$$

Now compute the outer integral

$$\frac{Fh}{D} = \int_0^1 I(y)\,\mathrm{d}y = c_0 \left[\frac{1}{4}y + \frac{3}{4}y^2\right]_0^1$$

$$= c_0 \times 1 = 10^4\ [\mathrm{mol/cm}].$$

Therefore,

$$D = \frac{Fh}{c_0} = 10^{-5}\ [\mathrm{cm}^2/\mathrm{s}].$$

Now consider the intended setup where $c(x,y) = c_0$, the integration becomes

$$\int_0^1 \int_0^1 c_0\,\mathrm{d}x\,\mathrm{d}y = c_0 \int_0^1 [x]_0^1\,\mathrm{d}y$$

$$= c_0 [y]_0^1 = c_0 = 10^4\ [\mathrm{mol/cm}].$$

Solution (cont)

and so

$$D = \frac{Fh}{c_0} = 10^{-5}\ [\mathrm{cm}^2/\mathrm{s}].$$

So, we see that in both cases the result is the same, the researchers got lucky!

Recap

Key points from section 3.5

- The double integral

$$\int_a^b \int_c^d f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

  consists of
  - an **inner integral**: evaluated first to give

$$\int_c^d f(x,y)\,\mathrm{d}x = g(y)$$

  - and an **outer integral**: evaluated using the result

$$\int_a^b g(y)\,\mathrm{d}y$$

- The result is the **volume** between the surface $z = f(x,y)$ and the plane $z = 0$ in the **rectangular region** $a \leq y \leq b$; $c \leq x \leq d$.

4 Vectors

Intended Learning Outcomes

The relevant ILOs for this section are

- **[ILO5]** Calculate sums, differences and products involving vectors and use these to solve geometrical problems.

4.1 Introduction to vectors

Learning Outcomes

On completion of this subsection, students will be able to

- Identify vector quantities and scalar quantities
- Define a position vector and vectors between points
- Calculate the magnitude of vectors and normalise them

Figure 4.1 shows an electrolysis experiment with several quantities in the chamber highlighted. Some of these are represented by a **single number**, such as water **temperature** and **density**, or the **concentration** of Sodium ions and Chloride ions. These are are called **scalar** quantities.

The sodium ions experience a **force** to the right. The chloride ions experience a force to the left. These forces give rise to **acceleration**. These have both **magnitude** and **direction**, and are called **vectors**.

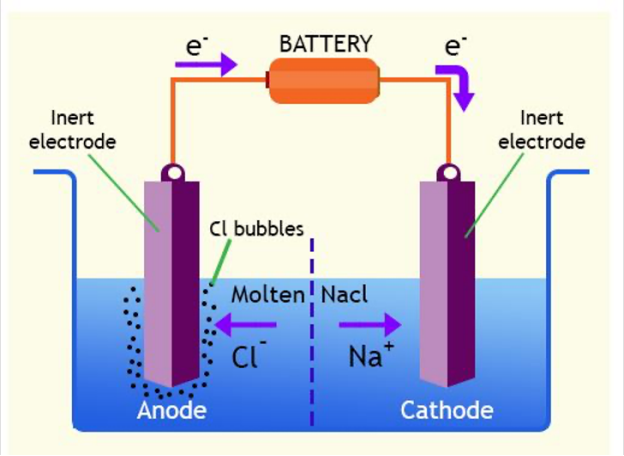


Figure 4.1: A sketch of an electrolysis chamber.

Some example scalar and vector quantities are given in 5.

A vector is an entity giving the position of one point with respect to another. It is used to describe a physical quantity with a direction *e.g.* 2 km north of here, 5 m s^{-1} downwards. A vector has both

Scalars	Vectors
Mass	Force
Volume	Acceleration
Density	Momentum
Energy	Concentration gradient
Temperature	Temperature gradient
Speed	Velocity
Distance	Displacement
Concentration	

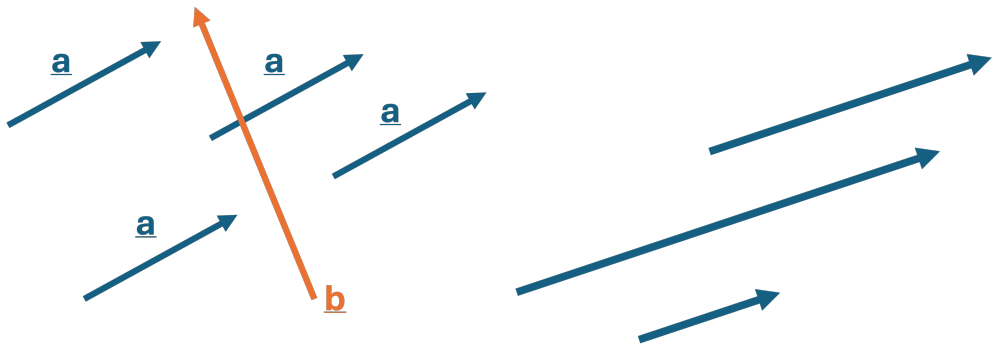
Table 5: Some examples of scalars and vector quantities.

magnitude and **direction**.

Notation.These notes will use bold and underlined letters (*e.g.* $\underline{\mathbf{a}}$) for vectors. The vector from point A to point B is written as $\overrightarrow{AB}$. When handwritten, typically vectors are written as $\underline{a}$ or $\overrightarrow{a}$. When typed, often they are just written in bold e.g. $\underline{\mathbf{a}}$.

A vector is characterised by its magnitude (length) and direction, not by the position of the head and tail. In figure 4.2(a) vectors $\underline{\mathbf{a}}$ are all the same vector but $\underline{\mathbf{b}}$ is a different vector.

Two vectors are said to be **parallel** if they point in the same direction, even if the magnitudes (lengths) are different. The three vectors shown in 4.2(b) are parallel.



(a) $\underline{\mathbf{a}}$ are all the same vector. (b) These vectors are parallel.

Figure 4.2

The vectors $\underline{\mathbf{i}}$, $\underline{\mathbf{j}}$, and $\underline{\mathbf{k}}$ are vectors of length 1 (*i.e.***unit vectors**) that point in the positive x, y, and z directions (figure 4.3(a)). A vector can be expressed in terms of $\underline{\mathbf{i}}$, $\underline{\mathbf{j}}$, and $\underline{\mathbf{k}}$. Note, sometimes, unit vectors are written with a "hat" e.g. $\underline{\hat{\mathbf{k}}}$.

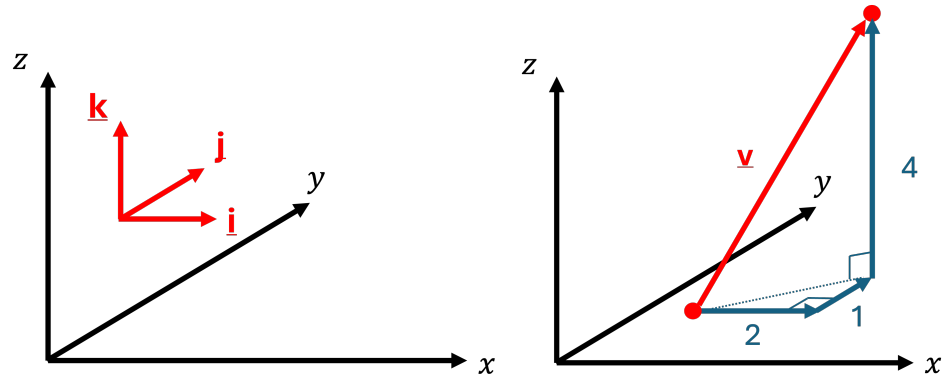
A vector representing a displacement of

- a units in the x-direction
- b units in the y-direction

- c units in the z-direction

can be represented as $a\underline{\mathbf{i}} + b\underline{\mathbf{j}} + c\underline{\mathbf{k}}$. For example, the vector $\underline{\mathbf{v}} = 2\underline{\mathbf{i}} + \underline{\mathbf{j}} + 4\underline{\mathbf{k}}$ (shown in figure 4.3(b)) goes

- 2 units to the right,
- 1 unit back (into the page),
- 4 units up.



(a) Sketch of the $\underline{\mathbf{i}}$, $\underline{\mathbf{j}}$, $\underline{\mathbf{k}}$ vectors. (b) The vector $\underline{\mathbf{v}} = 2\underline{\mathbf{i}} + \underline{\mathbf{j}} + 4\underline{\mathbf{k}}$

Figure 4.3

Alternatively, we can denote this as

$$\underline{\mathbf{v}} = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}.$$

Position Vector. The vector from the origin to a point is referred to as the **position vector** of that point. For example, the point $P = (4, 3)$ has position vector $\underline{\mathbf{p}} = 4\underline{\mathbf{i}} + 3\underline{\mathbf{j}}$ (figure 4.4(a)).

Vector from point to point. The vector from the point A with coordinates (a_x, a_y, a_z) to a point B with coordinates (b_x, b_y, b_z) is given by

$$\overrightarrow{AB} = (b_x - a_x)\underline{\mathbf{i}} + (b_y - a_y)\underline{\mathbf{j}} + (b_z - a_z)\underline{\mathbf{k}} = \underline{\mathbf{b}} - \underline{\mathbf{a}}. \tag{4.1.1}$$

E.g. the vector from $(2, 6, -1)$ to $(3, 4, 8)$ is

$$(3 - 2)\underline{\mathbf{i}} + (4 - 6)\underline{\mathbf{j}} + (8 + 1)\underline{\mathbf{k}} = \underline{\mathbf{i}} - 2\underline{\mathbf{j}} + 9\underline{\mathbf{k}}.$$

This is the same in two dimensions. E.g. vector from $A = (2, 6)$ to $B = (3, 4)$ is $\overrightarrow{AB} = \underline{\mathbf{i}} - 2\underline{\mathbf{j}}$ ((figure 4.4(b))).

Magnitude of a Vector. The magnitude, or length, of a vector, such as $\underline{\mathbf{a}} = a_x\underline{\mathbf{i}} + a_y\underline{\mathbf{j}} + a_z\underline{\mathbf{k}}$ is given

by

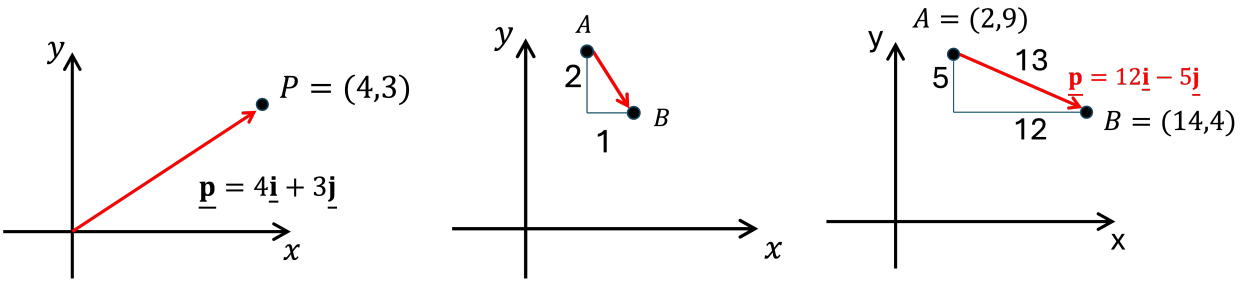
$$|\underline{\mathbf{a}}| = \sqrt{a_x^2 + a_y^2 + a_z^2}. \tag{4.1.2}$$

E.g. the magnitude of $\underline{\mathbf{b}} = 4\underline{\mathbf{i}} + \underline{\mathbf{j}} - 8\underline{\mathbf{k}}$ is

$$|\underline{\mathbf{b}}| = \sqrt{4^2 + 1^2 + (-8)^2} = \sqrt{16 + 1 + 64} = \sqrt{81} = 9.$$

In two dimensions, the formula is the same (with $a_z = 0$). For example, the vector from $A = (2, 9)$ to $B = (14, 4)$ is given by $\underline{\mathbf{p}} = 12\underline{\mathbf{i}} - 5\underline{\mathbf{j}}$ ((figure 4.4(c))). The magnitude of $\underline{\mathbf{p}}$ is

$$|\underline{\mathbf{p}}| = \sqrt{12^2 + 5^2} = \sqrt{169} = 13.$$



(a) Position vector (b) Point-to-point vector (c) Magnitude of a vector

Figure 4.4: Examples of various vector definitions

Vector scaling and normalisation. Suppose we have a vector $\underline{\mathbf{a}}$ and we want to find a vector $\underline{\mathbf{b}}$ with the same direction as $\underline{\mathbf{a}}$ but with a given magnitude m. Then

$$\underline{\mathbf{b}} = m\frac{\underline{\mathbf{a}}}{|\underline{\mathbf{a}}|}. \tag{4.1.3}$$

A special case is "normalisation" of $\underline{\mathbf{a}}$, where we find the "unit vector" parallel to $\underline{\mathbf{a}}$, given by

$$\hat{\underline{\mathbf{a}}} = \frac{1}{|\underline{\mathbf{a}}|}\underline{\mathbf{a}}. \tag{4.1.4}$$

The vector $\hat{\underline{\mathbf{a}}}$ has the same direction as $\underline{\mathbf{a}}$, but with magnitude 1.

Example 4.1.1 (Lecture). Given the points $A = (4, 2, 11)$ and $B = (6, 5, 5)$ find

- The position vector $\underline{\mathbf{a}}$ of the point A
- The vector $\overrightarrow{AB}$
- The magnitude of vector $\overrightarrow{AB}$
- The vector $\underline{\mathbf{v}}$ parallel to $\overrightarrow{AB}$ with magnitude 2

Solution

- $\underline{\mathbf{a}} = 4\underline{\mathbf{i}} + 2\underline{\mathbf{j}} + 11\underline{\mathbf{k}}$.
- $\overrightarrow{AB} = \underline{\mathbf{b}} - \underline{\mathbf{a}} = (6-4)\underline{\mathbf{i}} + (5-2)\underline{\mathbf{j}} + (5-11)\underline{\mathbf{k}} = 2\underline{\mathbf{i}} + 3\underline{\mathbf{j}} - 6\underline{\mathbf{k}}$.
- $|\overrightarrow{AB}| = \sqrt{2^2 + 3^2 + (-6)^2} = \sqrt{4 + 9 + 36} = 7$.
- $\underline{\mathbf{v}} = 2\frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = \frac{2}{7}\left(2\underline{\mathbf{i}} + 3\underline{\mathbf{j}} - 6\underline{\mathbf{k}}\right) = \frac{4}{7}\underline{\mathbf{i}} + \frac{6}{7}\underline{\mathbf{j}} - \frac{12}{7}\underline{\mathbf{k}}$.

Recap

Key points from section 4.1

- **Vectors**, unlike scalars, have a **direction** as well as a magnitude.
- Vectors can be written in the form

$$a\underline{\mathbf{i}} + b\underline{\mathbf{j}} + c\underline{\mathbf{k}} \qquad \text{or} \qquad \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

- The position vector of a point $P = (x, y, z)$ is

$$\underline{\mathbf{p}} = x\underline{\mathbf{i}} + y\underline{\mathbf{j}} + z\underline{\mathbf{k}}.$$

- To calculate the vector from A to B, subtract the coordinates of A from those of B.

4.2 Additive operations on vectors

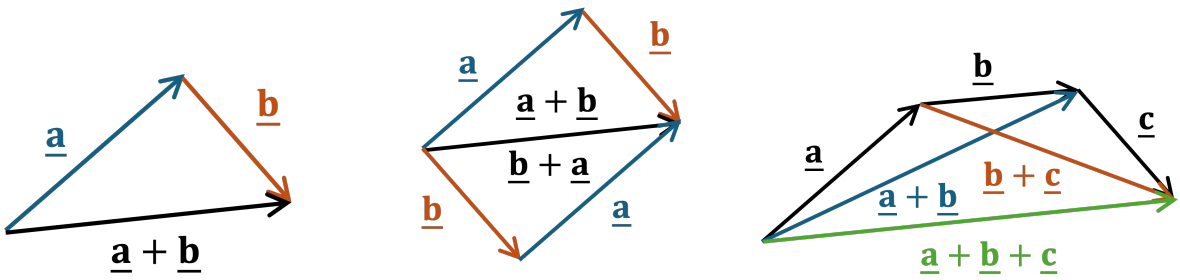
Learning Outcomes

On completion of this subsection, students will be able to

- Perform addition and subtraction of vector quantities
- Multiply vectors by scalar quantities

Addition of Vectors. The sum of two vectors is another vector. To find $\underline{\mathbf{a}} + \underline{\mathbf{b}}$, put the 'tail' of $\underline{\mathbf{b}}$ next to the head of $\underline{\mathbf{a}}$. Then $\underline{\mathbf{a}} + \underline{\mathbf{b}}$ joins the 'tail' of $\underline{\mathbf{a}}$ to the head of $\underline{\mathbf{b}}$ (figure 4.5(a)).

Note that $\underline{\mathbf{a}} + \underline{\mathbf{b}}$ and $\underline{\mathbf{b}} + \underline{\mathbf{a}}$ are equal: *i.e.* addition of vectors is **commutative** (figure 4.5(b)). Furthermore, $(\underline{\mathbf{a}} + \underline{\mathbf{b}}) + \underline{\mathbf{c}}$ and $\underline{\mathbf{a}} + (\underline{\mathbf{b}} + \underline{\mathbf{c}})$ are equal: *i.e.* addition of vectors is **associative** (figure 4.5(c)).



(a) Vector addition. (b) Commutative property. (c) Associative property.

Figure 4.5: Geometric representations of vector addition.

To add two vectors, simply add the components. Take $\underline{\mathbf{a}} = 2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}}$ and $\underline{\mathbf{b}} = 3\underline{\mathbf{i}} + 2\underline{\mathbf{j}} - 4\underline{\mathbf{k}}$. Then,

$$\begin{aligned}\underline{\mathbf{a}} + \underline{\mathbf{b}} &= (2+3)\underline{\mathbf{i}} + ((-1)+2)\underline{\mathbf{j}} + (3+(-4))\underline{\mathbf{k}} \\ &= 5\underline{\mathbf{i}} + \underline{\mathbf{j}} - \underline{\mathbf{k}}.\end{aligned}$$

Subtraction of Vectors. The difference between vectors $\underline{\mathbf{a}} - \underline{\mathbf{b}}$, is also a vector. To find $\underline{\mathbf{a}} - \underline{\mathbf{b}}$, put the 'tails' next to each other and join the head of $\underline{\mathbf{b}}$ to the head of $\underline{\mathbf{a}}$ (figure 4.6).

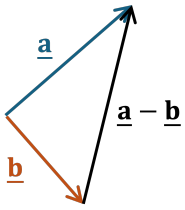


Figure 4.6: Geometric representation of vector subtraction

To subtract the vectors, simply subtract the components. Take $\underline{\mathbf{a}} = 2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}}$ and $\underline{\mathbf{b}} = 3\underline{\mathbf{i}} + 2\underline{\mathbf{j}} - 4\underline{\mathbf{k}}$. Then,

$$\begin{aligned}\underline{\mathbf{a}} - \underline{\mathbf{b}} &= (2-3)\underline{\mathbf{i}} + ((-1)-2)\underline{\mathbf{j}} + (3-(-4))\underline{\mathbf{k}} \\ &= -\underline{\mathbf{i}} - 3\underline{\mathbf{j}} + 7\underline{\mathbf{k}}.\end{aligned}$$

Multiplication of Vectors by Scalars. A vector $\underline{\mathbf{a}}$ can be multiplied by a scalar λ (a number). The result is a vector, *e.g.* $\underline{\mathbf{b}} = \lambda\underline{\mathbf{a}}$.

- For positive λ the result has the same direction as the original and the magnitude changes.
- For negative λ, the result points in the opposite direction (figure 4.7).

The resulting vector has all of its components multiplied by λ. So, if $\underline{\mathbf{a}} = a_x\underline{\mathbf{i}} + a_y\underline{\mathbf{j}} + a_z\underline{\mathbf{k}}$ then

$$\lambda\underline{\mathbf{a}} = \lambda a_x\underline{\mathbf{i}} + \lambda a_y\underline{\mathbf{j}} + \lambda a_z\underline{\mathbf{k}}\,. \tag{4.2.1}$$

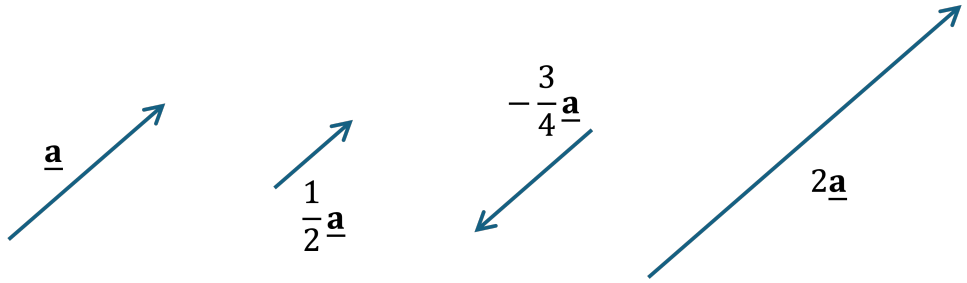


Figure 4.7: Different scalar multiples of the vector $\underline{\mathbf{a}}$.

E.g. Take $\underline{\mathbf{a}} = 2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}}$ and $\lambda = 5$, then

$$\begin{aligned}\underline{\mathbf{b}} = \lambda\underline{\mathbf{a}} &= 5(2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}}) \\ &= 10\underline{\mathbf{i}} - 5\underline{\mathbf{j}} + 15\underline{\mathbf{k}}.\end{aligned}$$

Distributive Laws. Note the distributive properties of multiplication by a scalar.

$$\lambda\underline{\mathbf{a}} + \lambda\underline{\mathbf{b}} = \lambda(\underline{\mathbf{a}} + \underline{\mathbf{b}}), \tag{4.2.2}$$

$$\lambda\underline{\mathbf{a}} + \mu\underline{\mathbf{a}} = (\lambda + \mu)\underline{\mathbf{a}}. \tag{4.2.3}$$

Example 4.2.1 (Lecture)**.** Calculate the vectors

$$\underline{\mathbf{u}} = 3\underline{\mathbf{a}} + 5\underline{\mathbf{a}}$$

and

$$\underline{\mathbf{v}} = \frac{1}{4}\underline{\mathbf{u}} + 2\underline{\mathbf{b}},$$

where

$$\underline{\mathbf{a}} = \underline{\mathbf{i}} - 2\underline{\mathbf{k}} \qquad \text{and} \qquad \underline{\mathbf{b}} = \frac{1}{2}\underline{\mathbf{j}} + 3\underline{\mathbf{k}}.$$

Solution

$$\underline{\mathbf{u}} = 3\underline{\mathbf{a}} + 5\underline{\mathbf{a}} = 8\underline{\mathbf{a}} = 8\underline{\mathbf{i}} - 16\underline{\mathbf{k}}.$$

$$\underline{\mathbf{v}} = \frac{1}{4}\underline{\mathbf{u}} + 2\underline{\mathbf{b}} = \frac{1}{4}(8\underline{\mathbf{i}} - 16\underline{\mathbf{k}}) + 2\left(\frac{1}{2}\underline{\mathbf{j}} + 3\underline{\mathbf{k}}\right) \qquad = 2\underline{\mathbf{i}} + \underline{\mathbf{j}} + 2\underline{\mathbf{k}}.$$

Example 4.2.2 (Video)**.** A "Janus particle" converts chemical energy into motion to "swim" on a surface. In an experiment the (x, y) position is measured every 10 seconds. One particle starts at $A = (2, 3)$ [μm], at the next measurement it is at $B = (5, 2)$ [μm].

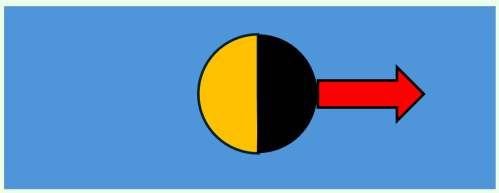
(a) Assuming it travels in a straight line, what is its velocity the first 10s of the experiment?

(b) The particle changes direction, travelling for the next 10s with average velocity

$$\underline{\mathbf{v}} = -0.1\underline{\mathbf{i}} + 0.4\underline{\mathbf{j}}\ \mu\text{m/s}.$$

What is its position at the next measurement?

(c) It continues to travel at the same velocity for the next 20s, how far is it from its starting point (40s into the experiment)?



Solution

(a) Define position vectors $\underline{\mathbf{a}} = 2\underline{\mathbf{i}} + 3\underline{\mathbf{j}}$ and $\underline{\mathbf{b}} = 5\underline{\mathbf{i}} + 2\underline{\mathbf{j}}$.
Velocity = displacement/time. The displacement is

$$\overrightarrow{AB} = (5-2)\underline{\mathbf{i}} + (2-3)\underline{\mathbf{j}} = 3\underline{\mathbf{i}} - \underline{\mathbf{j}} \quad [\mu\text{m}]$$

so velocity is $\underline{\mathbf{u}} = 0.3\underline{\mathbf{i}} - 0.1\underline{\mathbf{j}}$ [μm/s]

(b) $\underline{\mathbf{v}} = -0.1\underline{\mathbf{i}} + 0.4\underline{\mathbf{j}}$ mm/s so after 10s its position will be

$$\underline{\mathbf{c}} = \underline{\mathbf{b}} + 10\underline{\mathbf{v}} = (5-1)\underline{\mathbf{i}} + (2+4)\underline{\mathbf{j}} = 4\underline{\mathbf{i}} + 6\underline{\mathbf{j}} \quad [\mu\text{m}]$$

(c) After 20s more its position is

$$\underline{\mathbf{d}} = \underline{\mathbf{c}} + 20\underline{\mathbf{v}} = (4-3)\underline{\mathbf{i}} + (6+12)\underline{\mathbf{j}} = \underline{\mathbf{i}} + 18\underline{\mathbf{j}} \quad [\mu\text{m}]$$

The distance from the starting point is

$$|\overrightarrow{AD}| = |\underline{\mathbf{d}} - \underline{\mathbf{a}}| = |-\underline{\mathbf{i}} + 15\underline{\mathbf{j}}| = \sqrt{1+15^2} = \sqrt{226} \approx 15.03 \quad [\text{mm}]$$

Additive properties of vectors vs. scalars. Therefore, additive operations on vectors share all the same properties as additive operations on scalars, they are

- Commutative

- Associative
- Distributive

However, a **scalar and vector cannot be added together** directly, as they are fundamentally different objects.

Recap

Key points from section 4.2

- Vectors can be added / subtracted by **adding / subtracting their components**. The result is a vector.
- To multiply a vector by a scalar, **multiply each component** of the vector by the scalar. The result is a vector that is **(anti-)parallel to the original**.
- **Scalars cannot be added to vectors** directly, as they are fundamentally different objects.

4.3 The scalar product

Learning Outcomes

On completion of this subsection, students will be able to

- Define the scalar product of two vectors in terms of their magnitude and angle
- Calculate the scalar product of two vectors

The Scalar Product. Two vectors may be multiplied to give a scalar. The symbol $\cdot$ is used to denote this kind of multiplication. The scalar product of vectors $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ is written as

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}}.$$

It is often pronounced as $\underline{\mathbf{a}}$ 'dot' $\underline{\mathbf{b}}$ and is sometimes also referred to as the "dot product".

As the name suggests, the result of the scalar product is a **scalar**. Given two vectors, $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$, the scalar product $\underline{\mathbf{a}} \cdot \underline{\mathbf{b}}$ is defined as

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = |\underline{\mathbf{a}}||\underline{\mathbf{b}}| \cos\theta, \tag{4.3.1}$$

where θ is the angle between the vectors. It follows from the sign of $\cos\theta$ that if

- $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ are separated by an **acute** angle $(-\pi/2 < \theta < \pi/2)$, then $\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} > 0$;
- $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ are separated by an **obtuse** angle $(\pi/2 < |\theta| \leq \pi)$, then $\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} < 0$;
- $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ are separated by a **right** angle $(-\pi/2 < \theta < \pi/2)$, then $\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = 0$.

Two vectors at right angles are said to be **orthogonal** or **perpendicular**. The fact that the scalar product is zero is often used as a test of whether two vectors are perpendicular.

Conversely, if two vectors are parallel, then $\cos\theta = 1$ and so the scalar product is equal to the product of their magnitudes. A special case of this is the scalar product of a vector with itself

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{a}} = |\underline{\mathbf{a}}|^2.$$

Example 4.3.1 (Lecture)**.** Given the vectors

$$\underline{\mathbf{a}} = \underline{\mathbf{i}} + \underline{\mathbf{j}} \qquad \text{and} \qquad \underline{\mathbf{b}} = 5\underline{\mathbf{i}},$$

find the scalar product

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}}.$$

Solution

Calculate the scalar product using

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = |\underline{\mathbf{a}}||\underline{\mathbf{b}}| \cos\theta.$$

$$|\underline{\mathbf{a}}| = \sqrt{1^2 + 1^2} = \sqrt{2}, \qquad |\underline{\mathbf{b}}| = \sqrt{5^2} = 5.$$

By trig, the angle is given by

$$\theta = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}.$$

And so

$$\cos\theta = \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}.$$

Therefore,

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = \sqrt{2} \times 5 \times \frac{1}{\sqrt{2}} = 5.$$

Calculating the Scalar Product. An alternative, and more practical, formula for the scalar product is as follows. If $\underline{\mathbf{a}} = a_i\underline{\mathbf{i}} + a_j\underline{\mathbf{j}} + a_k\underline{\mathbf{k}}$ and $\underline{\mathbf{b}} = b_i\underline{\mathbf{i}} + b_j\underline{\mathbf{j}} + b_k\underline{\mathbf{k}}$, then

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = a_i b_i + a_j b_j + a_k b_k. \tag{4.3.2}$$

I.e. the sum of the products of each component.

For example, if $\underline{\mathbf{a}} = 2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}}$ and $\underline{\mathbf{b}} = 3\underline{\mathbf{i}} + 2\underline{\mathbf{j}} - 4\underline{\mathbf{k}}$, then

$$\begin{aligned} \underline{\mathbf{a}} \cdot \underline{\mathbf{b}} &= 2 \times 3 + (-1) \times 2 + 3 \times (-4) \\ &= -8. \end{aligned}$$

The two different formulae for the scalar product mean that it can be used to find the angle between two vectors

$$\cos\theta = \frac{\underline{\mathbf{a}} \cdot \underline{\mathbf{b}}}{|\underline{\mathbf{a}}||\underline{\mathbf{b}}|}. \tag{4.3.3}$$

Example 4.3.2 (DIY). Given the vectors

$$\underline{\mathbf{a}} = \underline{\mathbf{i}} - \underline{\mathbf{j}} - 2\underline{\mathbf{k}} \qquad \text{and} \qquad \underline{\mathbf{b}} = 2\underline{\mathbf{i}} + \underline{\mathbf{j}} - \underline{\mathbf{k}},$$

find the angle between $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$.

Solution

Use the formula

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = a_i b_i + a_j b_j + a_k b_k$$

to calculate the scalar product

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = 2 - 1 + 2 = 3.$$

Calculate the magnitude of each vector

$$|\underline{\mathbf{a}}| = \sqrt{1^2 + (-1)^2 + (-2)^2} = \sqrt{6},$$

$$|\underline{\mathbf{b}}| = \sqrt{2^2 + 1^2 + (-1)^2} = \sqrt{6}.$$

Using the scalar product definition

$$\cos\theta = \frac{\underline{\mathbf{a}} \cdot \underline{\mathbf{b}}}{|\underline{\mathbf{a}}||\underline{\mathbf{b}}|} = \frac{3}{\sqrt{6}\sqrt{6}} = \frac{1}{2},$$

and so

$$\theta = \frac{\pi}{3}.$$

Projection of vectors with the scalar product. Consider a unit vector $\hat{\underline{\mathbf{a}}}$ (*i.e.* $|\hat{\underline{\mathbf{a}}}| = 1$), then the scalar product

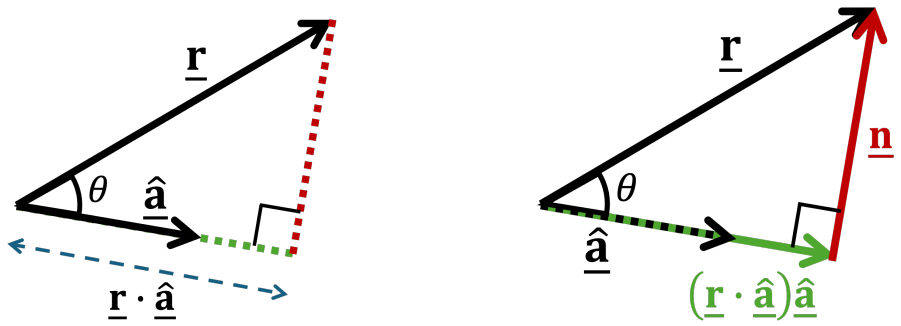
$$\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{a}}} = |\underline{\mathbf{r}}||\hat{\underline{\mathbf{a}}}|\cos\theta = |\underline{\mathbf{r}}|\cos\theta$$

is the component of $\underline{\mathbf{r}}$ that is in the direction of $\hat{\underline{\mathbf{a}}}$, see figure 4.8(a). Therefore, as shown in figure 4.8(b) the vector $(\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{a}}})\hat{\underline{\mathbf{a}}}$ is just $\underline{\mathbf{r}}$ projected onto $\hat{\underline{\mathbf{a}}}$. Thus, the vector $\underline{\mathbf{r}}$ can be expressed as

$$\underline{\mathbf{r}} = (\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{a}}})\hat{\underline{\mathbf{a}}} + \underline{\mathbf{n}},$$

where $\underline{\mathbf{n}}$ is perpendicular to $\hat{\underline{\mathbf{a}}}$ and so

$$\hat{\underline{\mathbf{a}}} \cdot \underline{\mathbf{n}} = 0.$$



(a) $\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{a}}} = |\underline{\mathbf{r}}| \cos\theta$ (b) $\underline{\mathbf{r}} = (\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{a}}})\hat{\underline{\mathbf{a}}} + \underline{\mathbf{n}}$

Figure 4.8: Sketch showing how $\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{a}}}$ can be used to project $\underline{\mathbf{r}}$ in to the $\hat{\underline{\mathbf{a}}}$) direction.

Example 4.3.3 (Video)**.** Given the vectors

$$\underline{\mathbf{a}} = 2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 2\underline{\mathbf{k}} \qquad \text{and} \qquad \underline{\mathbf{b}} = 6\underline{\mathbf{i}} + 12\underline{\mathbf{k}},$$

write $\underline{\mathbf{b}}$ in the form

$$\underline{\mathbf{b}} = \lambda\hat{\underline{\mathbf{a}}} + \mu\hat{\underline{\mathbf{n}}},$$

where $\hat{\underline{\mathbf{a}}}$ is the unit vector parallel to $\underline{\mathbf{a}}$ and $\hat{\underline{\mathbf{n}}}$ is a unit vector perpendicular to $\underline{\mathbf{a}}$.

Solution

First normalise $\underline{\mathbf{a}}$:

$$\hat{\underline{\mathbf{a}}} = \frac{\underline{\mathbf{a}}}{|\underline{\mathbf{a}}|} = \frac{2}{3}\underline{\mathbf{i}} - \frac{1}{3}\underline{\mathbf{j}} + \frac{2}{3}\underline{\mathbf{k}}$$

Then λ is given by the scalar product:

$$\lambda = \underline{\mathbf{b}} \cdot \hat{\underline{\mathbf{a}}} = 4 + 8 = 12$$

Subtracting the parallel part from $\underline{\mathbf{b}}$ leaves

$$\underline{\mathbf{n}} = \underline{\mathbf{b}} - \lambda\hat{\underline{\mathbf{a}}} = (6-8)\underline{\mathbf{i}} + 4\underline{\mathbf{j}} + (12-8)\underline{\mathbf{k}} = -2\underline{\mathbf{i}} + 4\underline{\mathbf{j}} + 4\underline{\mathbf{k}}$$

Comparing with the form in the question, $\underline{\mathbf{n}} = \mu\hat{\underline{\mathbf{n}}}$ and so

$$\mu = |\underline{\mathbf{n}}| = \sqrt{4 + 16 + 16} = 6$$

$$\hat{\underline{\mathbf{n}}} = \frac{\underline{\mathbf{n}}}{|\underline{\mathbf{n}}|} = -\frac{1}{3}\underline{\mathbf{i}} + \frac{2}{3}\underline{\mathbf{j}} + \frac{2}{3}\underline{\mathbf{k}}$$

And so $\underline{\mathbf{b}} = 12\hat{\underline{\mathbf{a}}} + 6\hat{\underline{\mathbf{n}}}$

Recap

Key points from section 4.3

- The scalar product of vectors $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ is defined as

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = |\underline{\mathbf{a}}||\underline{\mathbf{b}}| \cos\theta.$$

- It can also be calculated by summing the products of the vector components. The result is a **scalar** quantity.
- The angle between two vectors can be calculated by

$$\cos\theta = \frac{\underline{\mathbf{a}} \cdot \underline{\mathbf{b}}}{|\underline{\mathbf{a}}||\underline{\mathbf{b}}|}.$$

- The scalar product of a vector with itself is

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{a}} = |\underline{\mathbf{a}}|^2.$$

4.4 The vector product

Learning Outcomes

On completion of this subsection, students will be able to

- Calculate the vector product of two vectors and identify its direction from the directions of the two vectors.
- Use the vector product to calculate areas of parallelograms and triangles.

The vector product. Two vectors $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ may be multiplied together to form the vector product $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$, sometimes also referred to as the "cross product". As the name suggests, the result is a **vector**. Its magnitude is given by

$$|\underline{\mathbf{a}} \times \underline{\mathbf{b}}| = |\underline{\mathbf{a}}||\underline{\mathbf{b}}| \sin\theta, \tag{4.4.1}$$

where θ is the angle between $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$.

The vector product $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$ is perpendicular to both $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$. But how do we know whether $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$ should point up or down?

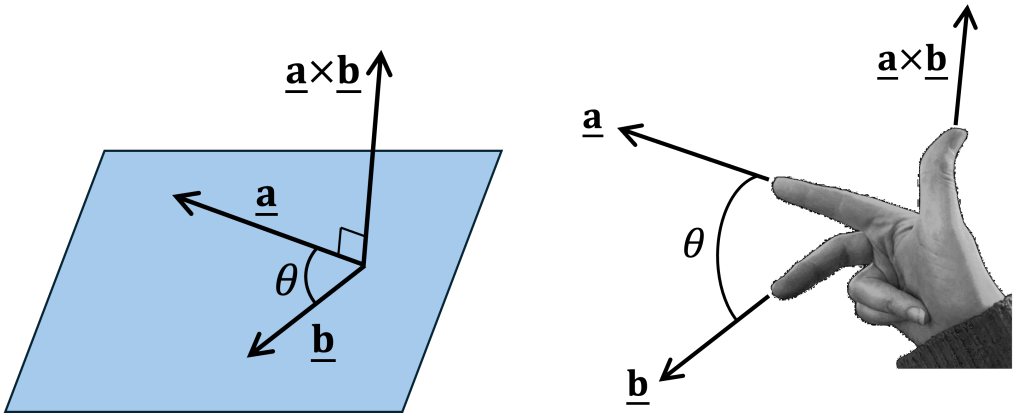
The vectors $\underline{\mathbf{a}}$, $\underline{\mathbf{b}}$ and $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$ form a right-handed system. Cartesian coordinates (x, y, z) also form a right-handed system, so:

$$\underline{\mathbf{i}} \times \underline{\mathbf{j}} = \underline{\mathbf{k}}, \qquad \underline{\mathbf{j}} \times \underline{\mathbf{k}} = \underline{\mathbf{i}}, \qquad \underline{\mathbf{k}} \times \underline{\mathbf{i}} = \underline{\mathbf{j}},$$

$$\underline{\mathbf{i}} \times \underline{\mathbf{k}} = -\underline{\mathbf{j}}, \qquad \underline{\mathbf{j}} \times \underline{\mathbf{i}} = -\underline{\mathbf{k}}, \qquad \underline{\mathbf{k}} \times \underline{\mathbf{j}} = -\underline{\mathbf{i}}.$$

There are several ways to remember what this means in practice, a common one is the so-called "right-hand rule".

The right-hand rule. Pointing your **index finger** along $\underline{\mathbf{a}}$ and directing your hand so that your **middle finger** points towards $\underline{\mathbf{b}}$. Then your thumb will point in the direction of $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$ (see figure 4.9). This is also visualised in the animation in figure 4.10.



(a) (b)

Figure 4.9: Demonstration of the right-hand rule, shown alongside the vector product $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$.

Example 4.4.1 (DIY)**.** A wire carries current $I = 1$ [A] along its length, given by the vector

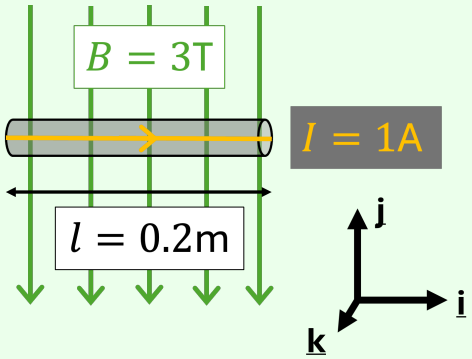
$$\underline{\mathbf{l}} = 0.2\underline{\mathbf{i}}\ [\mathrm{m}].$$

It is placed in a magnetic field

$$\underline{\mathbf{B}} = -3\underline{\mathbf{j}}\ [\mathrm{T}].$$

Calculate the magnetic force acting on the wire, given by

$$\underline{\mathbf{F}} = I(\underline{\mathbf{l}} \times \underline{\mathbf{B}}).$$



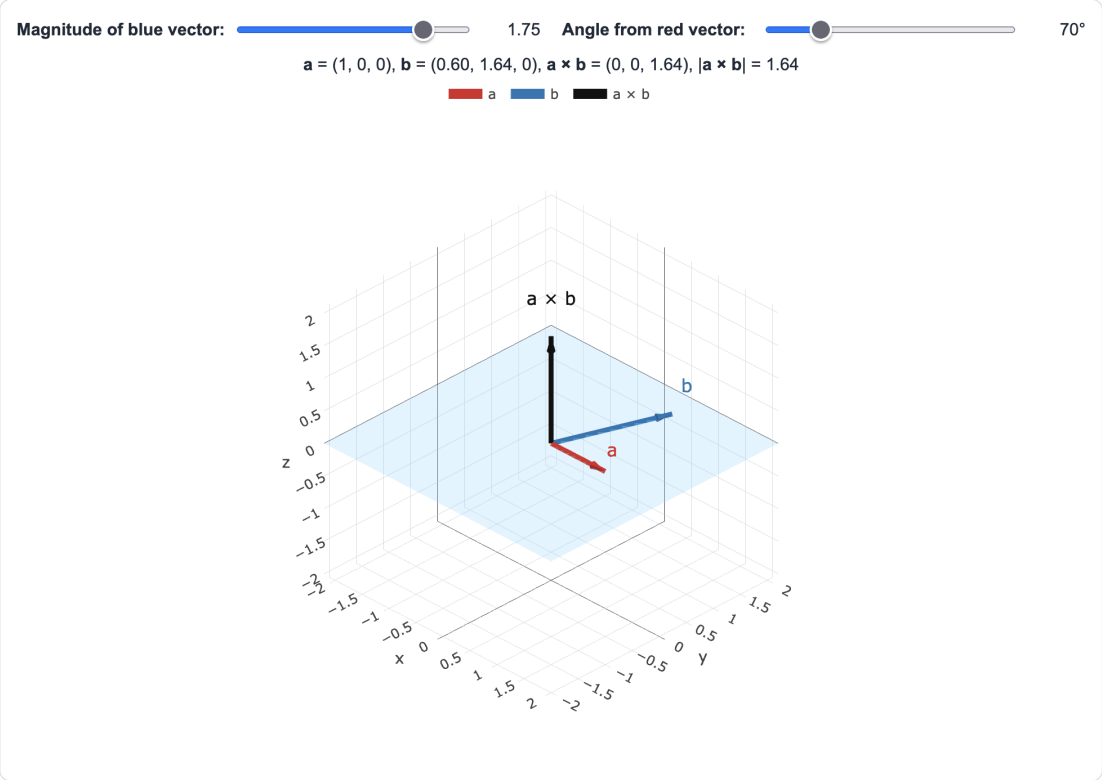


Figure 4.10: Screenshots of the interactive plot of the vector product $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$, available in the HTML version of these notes. The black vector show the resulting vector from taking the cross product of the red vector with the blue vector.

Solution

Using the right-hand rule, the direction of $\underline{\mathbf{l}} \times \underline{\mathbf{B}}$ is $-\underline{\mathbf{k}}$.
The magnitude of $\underline{\mathbf{l}} \times \underline{\mathbf{B}}$ is given by

$$|\underline{\mathbf{l}} \times \underline{\mathbf{B}}| = |\underline{\mathbf{l}}||\underline{\mathbf{B}}| \sin\theta = 0.2 \times 3 \times 1 = 0.6.$$

And so

$$\underline{\mathbf{F}} = 1 \times 0.6 \times (-\underline{\mathbf{k}}) = -0.6\underline{\mathbf{k}}\ [\mathrm{N}].$$

The Vector Product in Component form. Defining $\underline{\mathbf{a}} = (a_x\underline{\mathbf{i}} + a_y\underline{\mathbf{j}} + a_z\underline{\mathbf{k}})$ and $\underline{\mathbf{b}} = (b_x\underline{\mathbf{i}} + b_y\underline{\mathbf{j}} + b_z\underline{\mathbf{k}})$, the vector product is given by

$$\underline{\mathbf{a}} \times \underline{\mathbf{b}} = (a_y b_z - a_z b_y)\underline{\mathbf{i}} + (a_z b_x - a_x b_z)\underline{\mathbf{j}} + (a_x b_y - a_y b_x)\underline{\mathbf{k}}. \tag{4.4.2}$$

This is equivalent to the following determinant

$$\underline{\mathbf{a}} \times \underline{\mathbf{b}} = \begin{vmatrix} \underline{\mathbf{i}} & \underline{\mathbf{j}} & \underline{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}. \tag{4.4.3}$$

Example 4.4.2 (Lecture). If

$$\underline{\mathbf{a}} = 2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}} \qquad \text{and} \qquad \underline{\mathbf{b}} = 3\underline{\mathbf{i}} + 2\underline{\mathbf{j}} - 4\underline{\mathbf{k}},$$

find the vector product $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$.

Solution

$$\begin{aligned} \underline{\mathbf{a}} \times \underline{\mathbf{b}} &= \begin{vmatrix} \underline{\mathbf{i}} & \underline{\mathbf{j}} & \underline{\mathbf{k}} \\ 2 & -1 & 3 \\ 3 & 2 & -4 \end{vmatrix} \\ &= ((-1)(-4) - 3 \times 2)\underline{\mathbf{i}} + (3 \times 3 - 2(-4))\underline{\mathbf{j}} + (2 \times 2 - (-1)3)\underline{\mathbf{k}} \\ &= (4-6)\underline{\mathbf{i}} + (9+8)\underline{\mathbf{j}} + (4+3)\underline{\mathbf{k}} = -2\underline{\mathbf{i}} + 17\underline{\mathbf{j}} + 7\underline{\mathbf{k}} \end{aligned}$$

Properties of Vector Products. The vector product does not commute, in fact it is **anti-commutative**, so

$$\underline{\mathbf{a}} \times \underline{\mathbf{b}} = -(\underline{\mathbf{b}} \times \underline{\mathbf{a}}). \tag{4.4.4}$$

Multiplying vectors by a scalar only changes their magnitude, not direction, so

$$(\alpha\underline{\mathbf{a}}) \times \underline{\mathbf{b}} = \alpha(\underline{\mathbf{a}} \times \underline{\mathbf{b}}) = \underline{\mathbf{a}} \times (\alpha\underline{\mathbf{b}}). \tag{4.4.5}$$

This follows from the definition $|\underline{\mathbf{a}} \times \underline{\mathbf{b}}| = |\underline{\mathbf{a}}||\underline{\mathbf{b}}| \sin\theta$, since

$$|\alpha\underline{\mathbf{a}}||\underline{\mathbf{b}}| \sin\theta = \alpha|\underline{\mathbf{a}}||\underline{\mathbf{b}}| \sin\theta = |\underline{\mathbf{a}}||\alpha\underline{\mathbf{b}}| \sin\theta.$$

Example 4.4.3 (Video). For the vectors

$$\underline{\mathbf{a}} = 2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}} \qquad \text{and} \qquad \underline{\mathbf{b}} = 3\underline{\mathbf{i}} + 2\underline{\mathbf{j}} - 4\underline{\mathbf{k}}$$

and scalar $\alpha = 5$, verify that

$$(\alpha\underline{\mathbf{a}}) \times \underline{\mathbf{b}} = \underline{\mathbf{a}} \times (\alpha\underline{\mathbf{b}}) = \alpha(\underline{\mathbf{a}} \times \underline{\mathbf{b}}).$$

Solution

First calculate

$$\alpha\underline{\mathbf{a}} = 5(2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}}) = 10\underline{\mathbf{i}} - 5\underline{\mathbf{j}} + 15\underline{\mathbf{k}}$$
$$\alpha\underline{\mathbf{b}} = 5(3\underline{\mathbf{i}} + 2\underline{\mathbf{j}} - 4\underline{\mathbf{k}}) = 15\underline{\mathbf{i}} + 10\underline{\mathbf{j}} - 20\underline{\mathbf{k}}$$

Take the first case

$$(\alpha\underline{\mathbf{a}}) \times \underline{\mathbf{b}} = \begin{vmatrix} \underline{\mathbf{i}} & \underline{\mathbf{j}} & \underline{\mathbf{k}} \\ 10 & -5 & 15 \\ 3 & 2 & -4 \end{vmatrix} = (20-30)\underline{\mathbf{i}} + (45-(-40))\underline{\mathbf{j}} + (20-(-15))\underline{\mathbf{k}}$$
$$= -10\underline{\mathbf{i}} + 85\underline{\mathbf{j}} + 35\underline{\mathbf{k}}$$

In the second case

$$\underline{\mathbf{a}} \times (\alpha\underline{\mathbf{b}}) = \begin{vmatrix} \underline{\mathbf{i}} & \underline{\mathbf{j}} & \underline{\mathbf{k}} \\ 2 & -1 & 3 \\ 15 & 10 & -20 \end{vmatrix} = (20-30)\underline{\mathbf{i}} + (45-(-40))\underline{\mathbf{j}} + (20-(-15))\underline{\mathbf{k}}$$
$$= -10\underline{\mathbf{i}} + 85\underline{\mathbf{j}} + 35\underline{\mathbf{k}}$$

Finally

$$\alpha(\underline{\mathbf{a}} \times \underline{\mathbf{b}}) = 5\begin{vmatrix} \underline{\mathbf{i}} & \underline{\mathbf{j}} & \underline{\mathbf{k}} \\ 2 & -1 & 3 \\ 3 & 2 & -4 \end{vmatrix} = 5[(4-6)\underline{\mathbf{i}} + (9-(-8))\underline{\mathbf{j}} + (4-(-3))\underline{\mathbf{k}}]$$
$$= 5(-2\underline{\mathbf{i}} + 17\underline{\mathbf{j}} + 7\underline{\mathbf{k}}) = -10\underline{\mathbf{i}} + 85\underline{\mathbf{j}} + 35\underline{\mathbf{k}}$$

The vector product is **distributive**, *i.e.*

$$\underline{\mathbf{a}} \times (\underline{\mathbf{b}} + \underline{\mathbf{c}}) = (\underline{\mathbf{a}} \times \underline{\mathbf{b}}) + (\underline{\mathbf{a}} \times \underline{\mathbf{c}}). \tag{4.4.6}$$

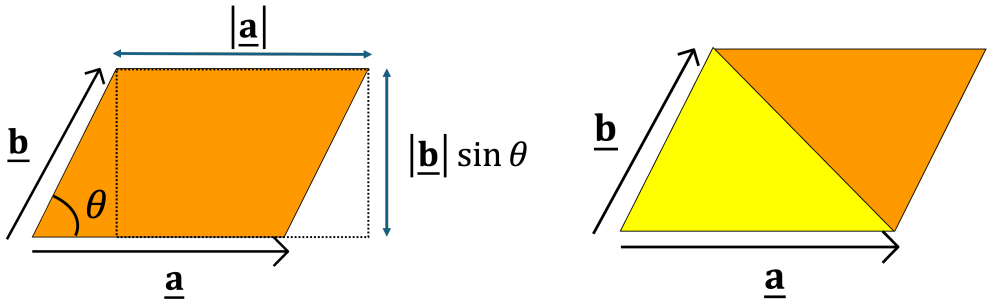
And so

$$\underline{\mathbf{a}} \times (\underline{\mathbf{b}} - \underline{\mathbf{c}}) = (\underline{\mathbf{a}} \times \underline{\mathbf{b}}) - (\underline{\mathbf{a}} \times \underline{\mathbf{c}}).$$

Parallel (and **anti-parallel**) vectors have a cross product of $\underline{\mathbf{0}}$. A special case of this is

$$\underline{\mathbf{a}} \times \underline{\mathbf{a}} = \underline{\mathbf{0}}.$$

Again, this can be seen from the definition $|\underline{\mathbf{a}} \times \underline{\mathbf{b}}| = |\underline{\mathbf{a}}||\underline{\mathbf{b}}| \sin\theta$, as the angle $\theta = 0$ for parallel vectors and $\theta = \pi$ for anti-parallel vectors.



(a) Parallelogram (b) Triangle

Figure 4.11: The magntiude of the vector product $|\underline{\mathbf{a}} \times \underline{\mathbf{b}}|$ gives the area of the parallelogram formed by $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ in (a). Halving this gives the area of the yellow (or orange) triangle in (b).

Areas of Parallelograms and Triangles. Consider the parallelogram with sides given by the vectors $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ in figure 4.11(a). The 'height' of the parallelogram is given by $|\underline{\mathbf{b}}| \sin\theta$. So, the area is

$$|\underline{\mathbf{a}}||\underline{\mathbf{b}}| \sin\theta = |\underline{\mathbf{a}} \times \underline{\mathbf{b}}| \,.$$

Now, consider the triangle with sides given by the vectors $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ in figure 4.11(b). The triangle has half of the area of the parallelogram in figure 4.11(a) and so the area is $\frac{1}{2}|\underline{\mathbf{a}} \times \underline{\mathbf{b}}|$.

Example 4.4.4 (Lecture)**.** Find the area of the triangle with two sides given by the vectors

$$\underline{\mathbf{a}} = 2\underline{\mathbf{i}} - \underline{\mathbf{j}} + 3\underline{\mathbf{k}} \qquad \text{and} \qquad \underline{\mathbf{b}} = -2\underline{\mathbf{i}} + 17\underline{\mathbf{j}} + 7\underline{\mathbf{k}}.$$

Solution

In determinant form

$$\begin{aligned}
\underline{\mathbf{a}} \times \underline{\mathbf{b}} &= \begin{vmatrix} \underline{\mathbf{i}} & \underline{\mathbf{j}} & \underline{\mathbf{k}} \\ 2 & -1 & 3 \\ -2 & 17 & 7 \end{vmatrix} \\
&= (-7-51)\underline{\mathbf{i}} + (-6-14)\underline{\mathbf{j}} + (34-2)\underline{\mathbf{k}} \\
&= -58\underline{\mathbf{i}} - 20\underline{\mathbf{j}} + 32\underline{\mathbf{k}}.
\end{aligned}$$

The area of the triangle is

$$\begin{aligned}
\frac{1}{2}|\underline{\mathbf{a}} \times \underline{\mathbf{b}}| &= \frac{1}{2}\sqrt{(-58)^2 + (-20)^2 + 32^2} \\
&= \frac{1}{2}\sqrt{4788} = \sqrt{1197} \approx 34.6.
\end{aligned}$$

⋆ Exam mistake of the week ⋆

Remember, the result of the **scalar** product is a **scalar**, and the result of the **vector** product is a **vector**.

25/26 In class assessment 2 (discontinued)

Given that

$$\underline{\mathbf{a}} = -8\underline{\mathbf{i}} - 2\underline{\mathbf{j}} + 4\underline{\mathbf{k}}$$

and that

$$\underline{\mathbf{b}} = 7\underline{\mathbf{i}} - 5\underline{\mathbf{j}} + 2\underline{\mathbf{k}},$$

input the scalar or "dot" product

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = \_\_\_\_\_.$$

Several people tried to input a vector, which was not accepted by STACK, causing some confusion and frustration.

Recap

Key points from section 4.4

- The result of $\underline{\mathbf{a}} \times \underline{\mathbf{b}}$ is a **vector** perpendicular to $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$.
- The direction of the result can be found by the **right-hand rule**.

  $$|\underline{\mathbf{a}} \times \underline{\mathbf{b}}| = |\underline{\mathbf{a}}||\underline{\mathbf{b}}| \sin\theta,$$

  which is the **area** of the relevant parallelogram.
- It can be calculated using the determinant, or in component form

  $$(a_y b_z - a_z b_y)\underline{\mathbf{i}} + (a_z b_x - a_x b_z)\underline{\mathbf{j}} + (a_x b_y - a_y b_x)\underline{\mathbf{k}}.$$

- The cross product of a vector with itself is

  $$\underline{\mathbf{a}} \times \underline{\mathbf{a}} = \underline{\mathbf{0}}.$$

4.5 Vectors, lines, and planes

Learning Outcomes

On completion of this subsection, students will be able to

- Write the vector equations for a line and a plane
- Find crossing points of lines and planes given in vector form

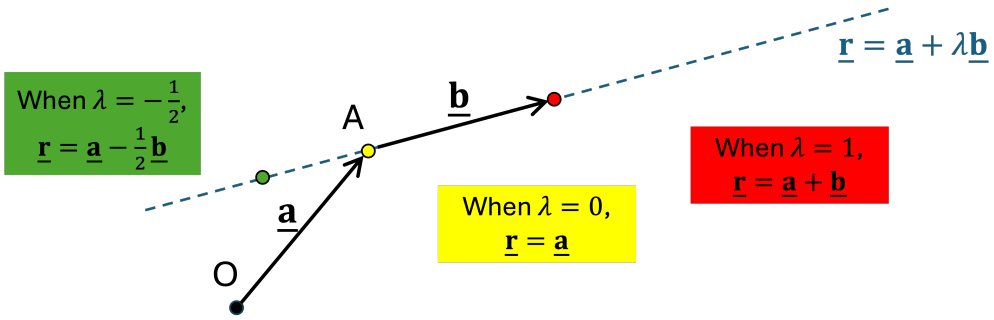


Figure 4.12: Illustration of points on the line given by different values of the parameter λ in equation (4.5.1).

Vectors and Lines. A line passing through point A (with position vector $\underline{\mathbf{a}}$) and parallel to vector $\underline{\mathbf{b}}$ has equation

$$\underline{\mathbf{r}} = \underline{\mathbf{a}} + \lambda\underline{\mathbf{b}}. \tag{4.5.1}$$

Here, λ is a parameter and every value of λ represents a different point on the line (see figure 4.12.

To find a line passing through points with position vectors $\underline{\mathbf{p}}$ and $\underline{\mathbf{q}}$:

- Choose $\lambda = 0$ for when $\underline{\mathbf{r}} = \underline{\mathbf{p}}$, *i.e.* set $\underline{\mathbf{a}} = \underline{\mathbf{p}}$.
- Choose $\lambda = 1$ for when $\underline{\mathbf{r}} = \underline{\mathbf{q}}$, *i.e.* set $\underline{\mathbf{b}} = \underline{\mathbf{q}} - \underline{\mathbf{p}}$.

Therefore, the equation becomes

$$\underline{\mathbf{r}} = \underline{\mathbf{p}} + \lambda(\underline{\mathbf{q}} - \underline{\mathbf{p}}).$$

Example 4.5.1 (Lecture)**.** (a) Find the vector equation of the line through the two points $(2, 1, 5)$ and $(4, 3, -2)$.
(b) By eliminating any parameter, write this in the form

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Solution

Define

$$\underline{\mathbf{p}} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} \quad \text{and} \quad \underline{\mathbf{q}} = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}.$$

Solution (cont)

Then the equation of the line is

$$\begin{aligned}\underline{\mathbf{r}} &= \underline{\mathbf{p}} + \lambda(\underline{\mathbf{q}} - \underline{\mathbf{p}}) \\ &= \begin{pmatrix}2\\1\\5\end{pmatrix} + \lambda \begin{pmatrix}2\\2\\-7\end{pmatrix}.\end{aligned}$$

(b) Writing

$$\underline{\mathbf{r}} = \begin{pmatrix}x\\y\\z\end{pmatrix}$$

gives equations for x, y, and z in terms of λ:

$$x = 2 + 2\lambda, \qquad y = 1 + 2\lambda, \qquad z = 5 - 7\lambda.$$

Each of these can be rearranged in terms of λ:

$$\lambda = \frac{x-2}{2}, \qquad \lambda = \frac{y-1}{2}, \qquad \lambda = \frac{z-5}{-7}.$$

All of these expressions must be equal, so

$$\frac{x-2}{2} = \frac{y-1}{2} = \frac{z-5}{-7}.$$

Comparing to the original expression:

$$x_0 = 2, \qquad a = 2, \qquad y_0 = 1, \qquad b = 2, \qquad z_0 = 5, \qquad c = -7.$$

To find whether the lines

$$\underline{\mathbf{r}}_1 = \underline{\mathbf{a}} + \lambda\underline{\mathbf{b}} \qquad \text{and} \qquad \underline{\mathbf{r}}_2 = \underline{\mathbf{c}} + \mu\underline{\mathbf{d}}$$

cross, we need to equate $\underline{\mathbf{r}}_1 = \underline{\mathbf{r}}_2$. *I.e.* it is necessary to find values of λ and μ that satisfy **all** components. If the lines do not cross, this will not be possible.

- In 2D, lines that do not cross are parallel, which would mean here that $\underline{\mathbf{b}}$ was parallel (or anti-parallel) to $\underline{\mathbf{d}}$ (see figure 4.13).
- In 3D, two lines do not cross in general.

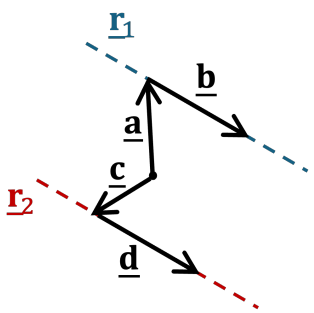


Figure 4.13: A sketch of when the lines $\underline{\mathbf{r}}_1 = \underline{\mathbf{a}} + \lambda\underline{\mathbf{b}}$ and $\underline{\mathbf{r}}_2 = \underline{\mathbf{c}} + \mu\underline{\mathbf{d}}$ are parallel.

Example 4.5.2 (Lecture)**.** There is one value of w for which the lines

$$\underline{\mathbf{r}}_1 = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

and

$$\underline{\mathbf{r}}_2 = \begin{pmatrix} 5 \\ 2 \\ w \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix}$$

cross. Find this value of w and the crossing point.

Solution

Start by setting the lines equal $\underline{\mathbf{r}}_1 = \underline{\mathbf{r}}_2$ and so the x- and y- components are

$$2 + \lambda = 5 + \mu, -1 + 2\lambda = 2 - \mu.$$

Adding these two equations together gives

$$1 + 3\lambda = 7$$

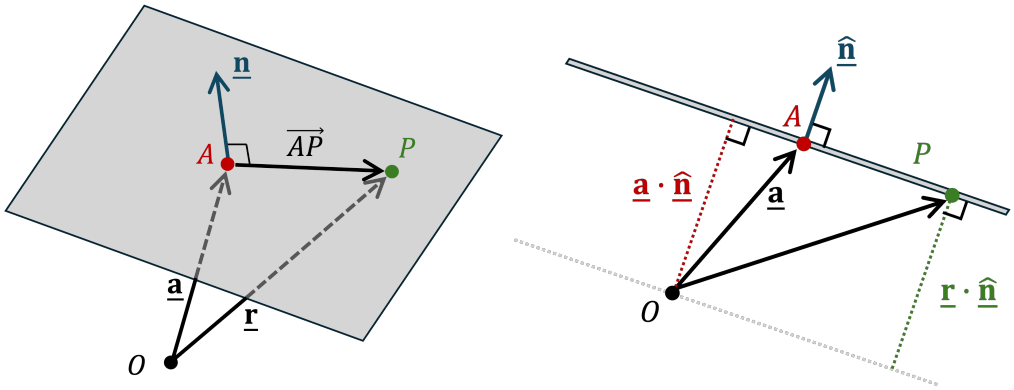
and so $\lambda = 2$ and therefore $\mu = -1$. Using these values, the z-component becomes

$$4 - 2 = w + 3.$$

Therefore, the lines cross when $w = -1$

Substitute the values in to find the crossing point, *e.g.* using $\lambda = 2$ in $\underline{\mathbf{r}}_1$ gives

$$\underline{\mathbf{r}}_1 = \begin{pmatrix} 2+2 \\ -1+4 \\ 4-2 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$$



(a) Looking down (b) Side view

Figure 4.14: Sketches of the definition of the plane defined by the point A and normal vector $\underline{\mathbf{n}}$. The equation $\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{n}}} = \underline{\mathbf{a}} \cdot \hat{\underline{\mathbf{n}}}$ is satisfied for a position vector $\underline{\mathbf{r}}$ that points to some point P on the plane.

Vector equation of a plane. A plane can be uniquely specified by

- A point on the plane, A (with position vector $\underline{\mathbf{a}}$).
- **and** a vector perpendicular to the plane, $\underline{\mathbf{n}}$.

If the point P (with position vector $\underline{\mathbf{r}}$) is also on the plane then $\overrightarrow{AP}$ must lie in the plane, and so $\overrightarrow{AP}$ is perpendicular to $\underline{\mathbf{n}}$ (figure 4.14(a)). Since $\overrightarrow{AP} = \underline{\mathbf{r}} - \underline{\mathbf{a}}$ is perpendicular to $\underline{\mathbf{n}}$, then $(\underline{\mathbf{r}} - \underline{\mathbf{a}}) \cdot \underline{\mathbf{n}} = 0$, and so, by the distributive property,

$$\underline{\mathbf{r}} \cdot \underline{\mathbf{n}} = \underline{\mathbf{a}} \cdot \underline{\mathbf{n}}.$$

This is the vector equation of a plane.

Defining

$$\underline{\mathbf{r}} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad \underline{\mathbf{n}} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \qquad \underline{\mathbf{a}} \cdot \underline{\mathbf{n}} = d,$$

we return to the algebraic form for the equation of a plane

$$ax + by + cz = d.$$

Note, often the unit normal vector $\hat{\underline{\mathbf{n}}}$ is used, so the equation of the plane becomes $\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{n}}} = \underline{\mathbf{a}} \cdot \hat{\underline{\mathbf{n}}}$.

An equivalent way to represent the equation of a plane is

$$\underline{\mathbf{r}} \cdot \hat{\underline{\mathbf{n}}} = d,$$

The geometric interpretation of this is that all points on the plane are the same perpendicular distance d from the origin, as sketched in figure 4.14(b).

Example 4.5.3 (Lecture). Write the vector equation of the plane that contains the points

$$A = (0, 2, -2), \qquad B = (6, 2, 0), \qquad C = (3, 5, 1).$$

Solution

$\overrightarrow{AB}$ and $\overrightarrow{AC}$ lie in the plane, therefore $\overrightarrow{AB} \times \overrightarrow{AC}$ is perpendicular to the plane

$$\overrightarrow{AB} = \begin{pmatrix} 6 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \\ 2 \end{pmatrix}, \qquad \overrightarrow{AC} = \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$

$$\begin{aligned} \overrightarrow{AB} \times \overrightarrow{AC} &= \begin{vmatrix} \underline{\mathbf{i}} & \underline{\mathbf{j}} & \underline{\mathbf{k}} \\ 6 & 0 & 2 \\ 3 & 3 & 3 \end{vmatrix} \\ &= -6\underline{\mathbf{i}} + (6 - 18)\underline{\mathbf{j}} + 18\underline{\mathbf{k}} = \begin{pmatrix} -6 \\ -12 \\ 18 \end{pmatrix} \\ &= \begin{pmatrix} -6 \\ -12 \\ 18 \end{pmatrix} \end{aligned}$$

Choose $\underline{\mathbf{n}} = -\frac{1}{6}(\overrightarrow{AB} \times \overrightarrow{AC}) = \underline{\mathbf{i}} + 2\underline{\mathbf{j}} - 3\underline{\mathbf{k}}$ and $\underline{\mathbf{a}} = 2\underline{\mathbf{j}} - 2\underline{\mathbf{k}}$ then

$$\underline{\mathbf{a}} \cdot \underline{\mathbf{n}} = 4 + 6 = 10$$

and so the equation of the plane is

$$\underline{\mathbf{r}} \cdot \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} = 10$$

Example 4.5.4 (Video). Find where the line

$$\underline{\mathbf{r}}_1 = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$$

intersects the plane

$$\underline{\mathbf{r}} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 5.$$

Solution

Substitute $\underline{\mathbf{r}} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$ into the equation for the plane

$$\begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 5$$

and so

$$1 + 2\lambda = 5$$

which gives $\lambda = 2$. The intersection point is found by substituting back into $\underline{\mathbf{r}}_1$

$$\begin{aligned} \underline{\mathbf{r}}_1(\lambda = 2) &= \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} 7 \\ 0 \\ -2 \end{pmatrix} = 7\underline{\mathbf{i}} - 2\underline{\mathbf{k}} \end{aligned}$$

Recap

Key points from section 4.5

- The vector equation of a line is
$$\underline{\mathbf{r}} = \underline{\mathbf{a}} + \lambda \underline{\mathbf{b}},$$
where $\underline{\mathbf{a}}$ is the position vector of a point on the line, $\underline{\mathbf{b}}$ gives its direction, and λ is a parameter.
- A line passing through points $\underline{\mathbf{p}}$ and $\underline{\mathbf{q}}$ is
$$\underline{\mathbf{r}} = \underline{\mathbf{p}} + \lambda(\underline{\mathbf{q}} - \underline{\mathbf{p}}).$$
- The vector equation of a plane is
$$\underline{\mathbf{r}} \cdot \underline{\mathbf{n}} = \underline{\mathbf{a}} \cdot \underline{\mathbf{n}},$$
where $\underline{\mathbf{a}}$ is the position vector of a point on the plane, and $\underline{\mathbf{n}}$ is normal to the plane.
- Crossing points of lines and planes can be found by equating the expressions for $\underline{\mathbf{r}}$.